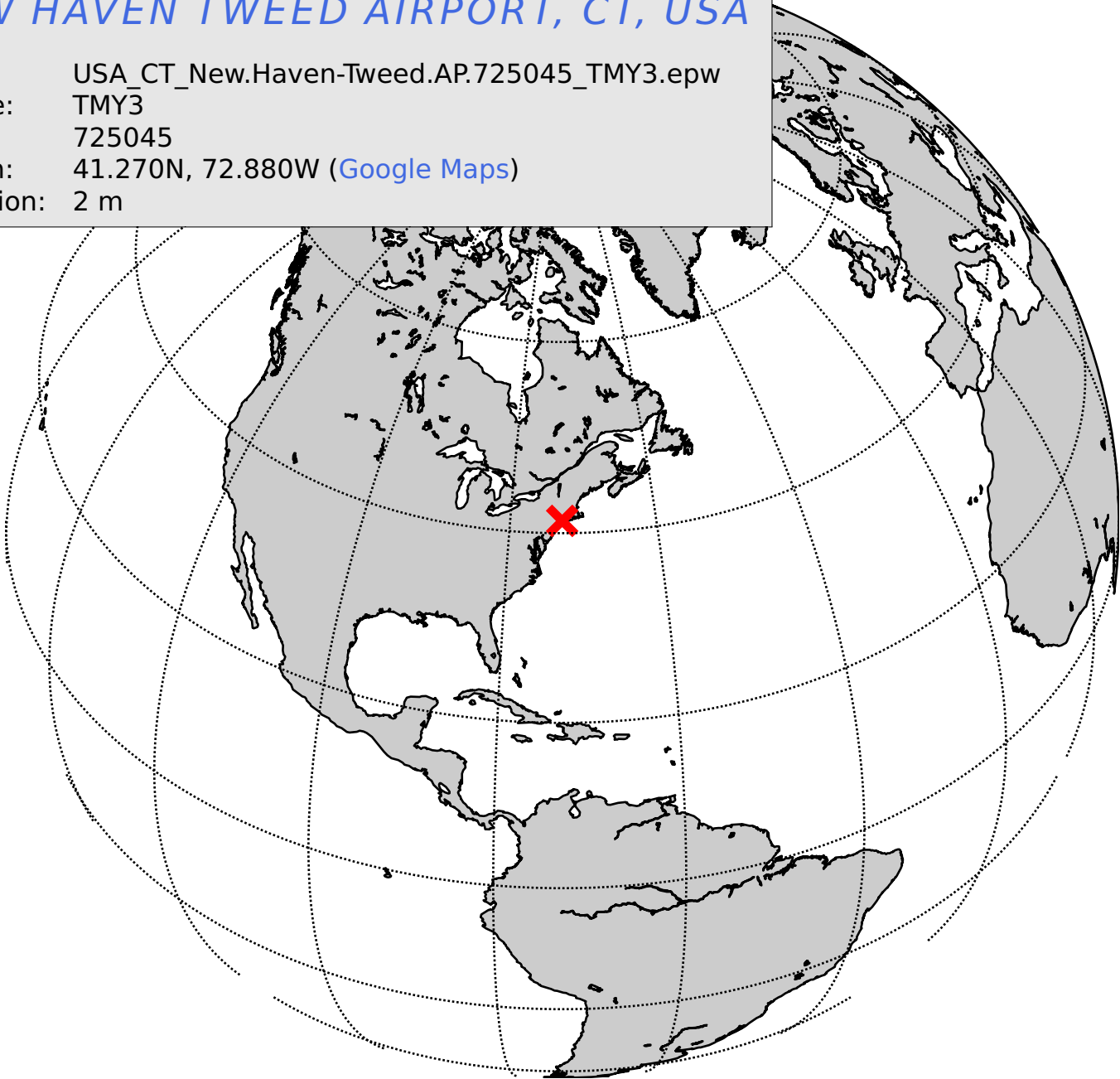


NEW HAVEN TWEED AIRPORT, CT, USA

From: USA_CT_New.Haven-Tweed.AP.725045_TMY3.epw
Source: TMY3
WMO: 725045
Lat/lon: 41.270N, 72.880W ([Google Maps](#))
Elevation: 2 m



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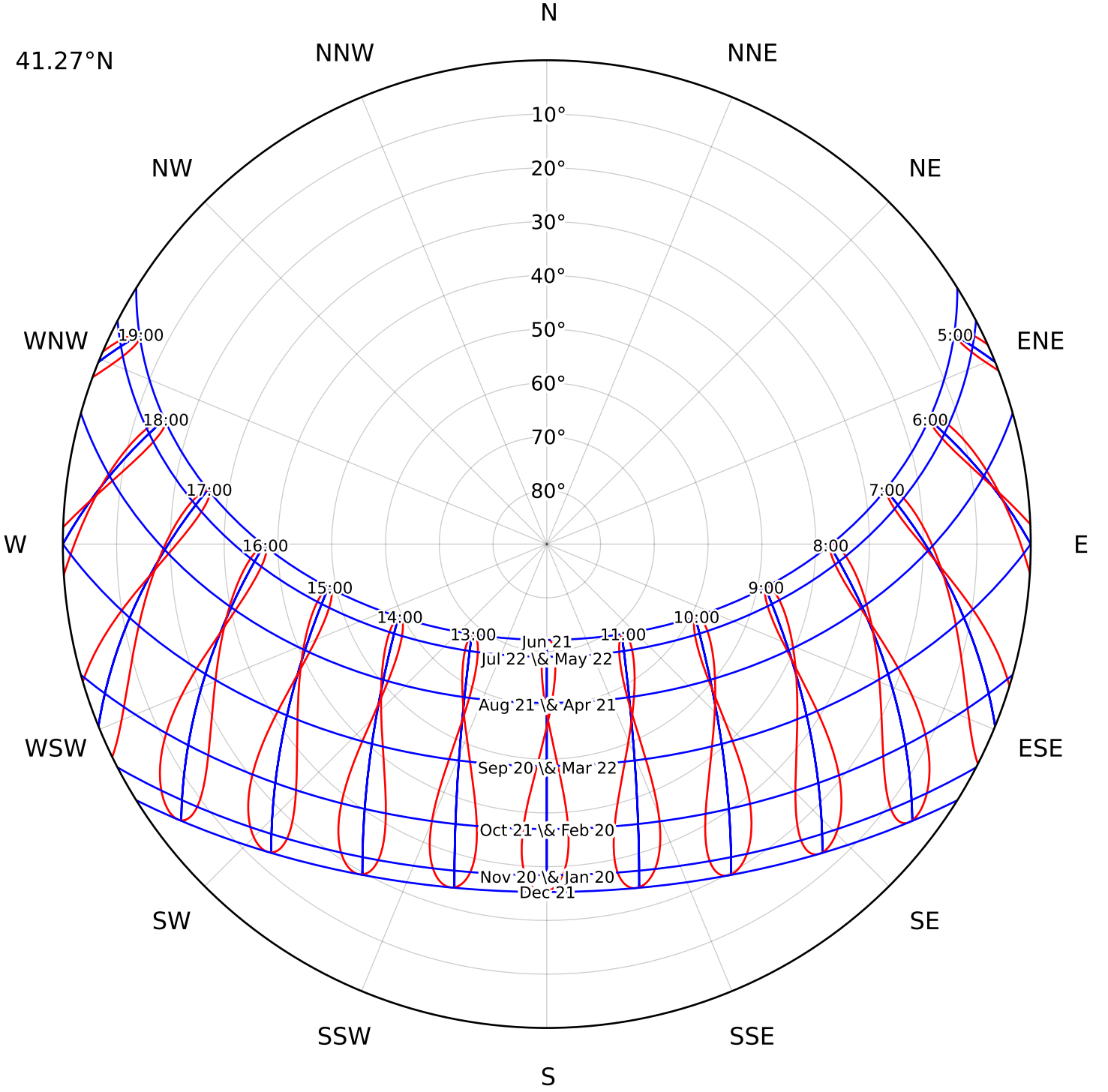


Figure 1: Path of sun at latitude of site. For a given hour of the day, the blue lines are based on solar time, while the red lines (analemma) are based on local time.

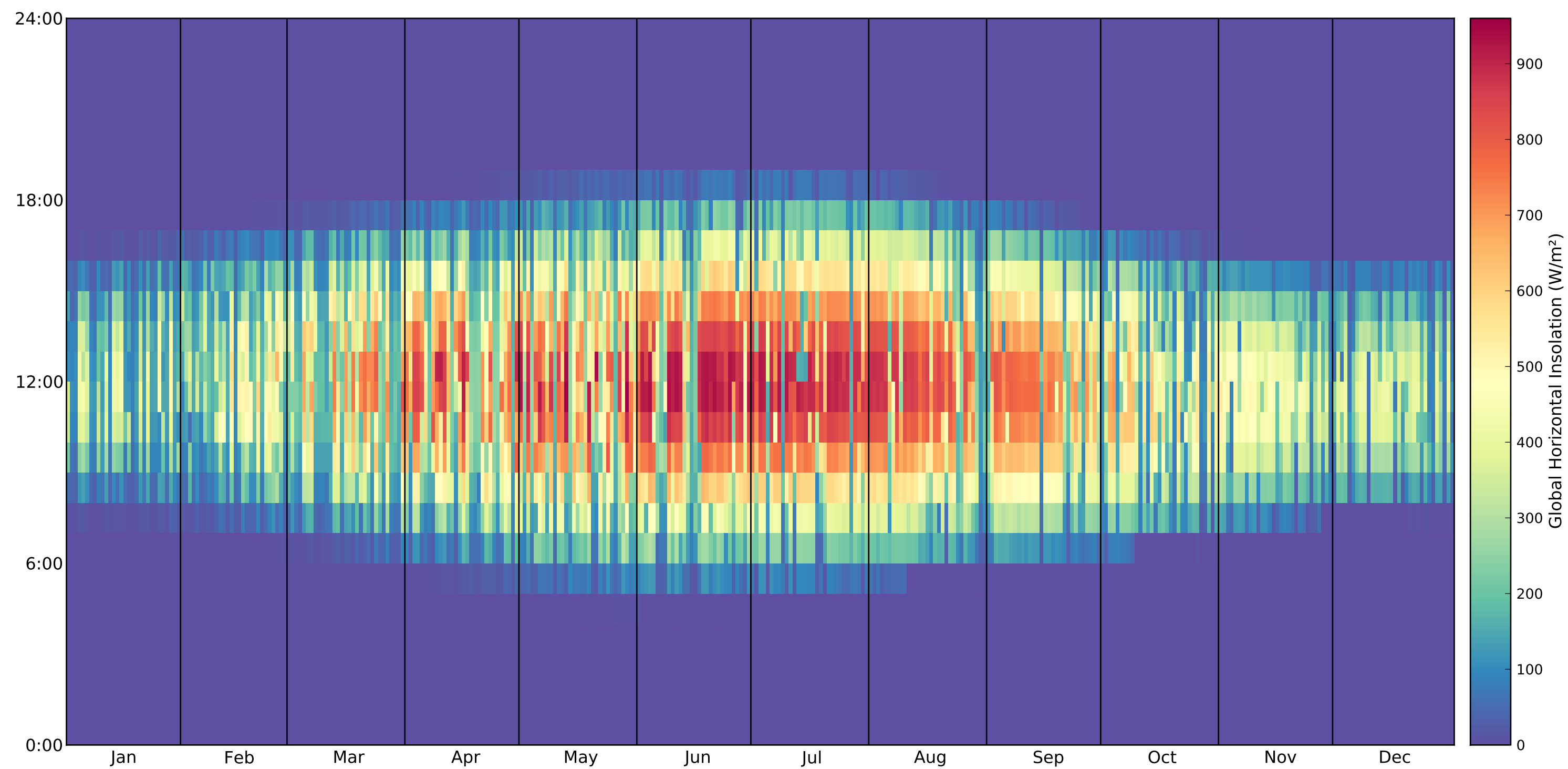


Figure 2: Global horizontal insolation for a given day of year and hour of day.

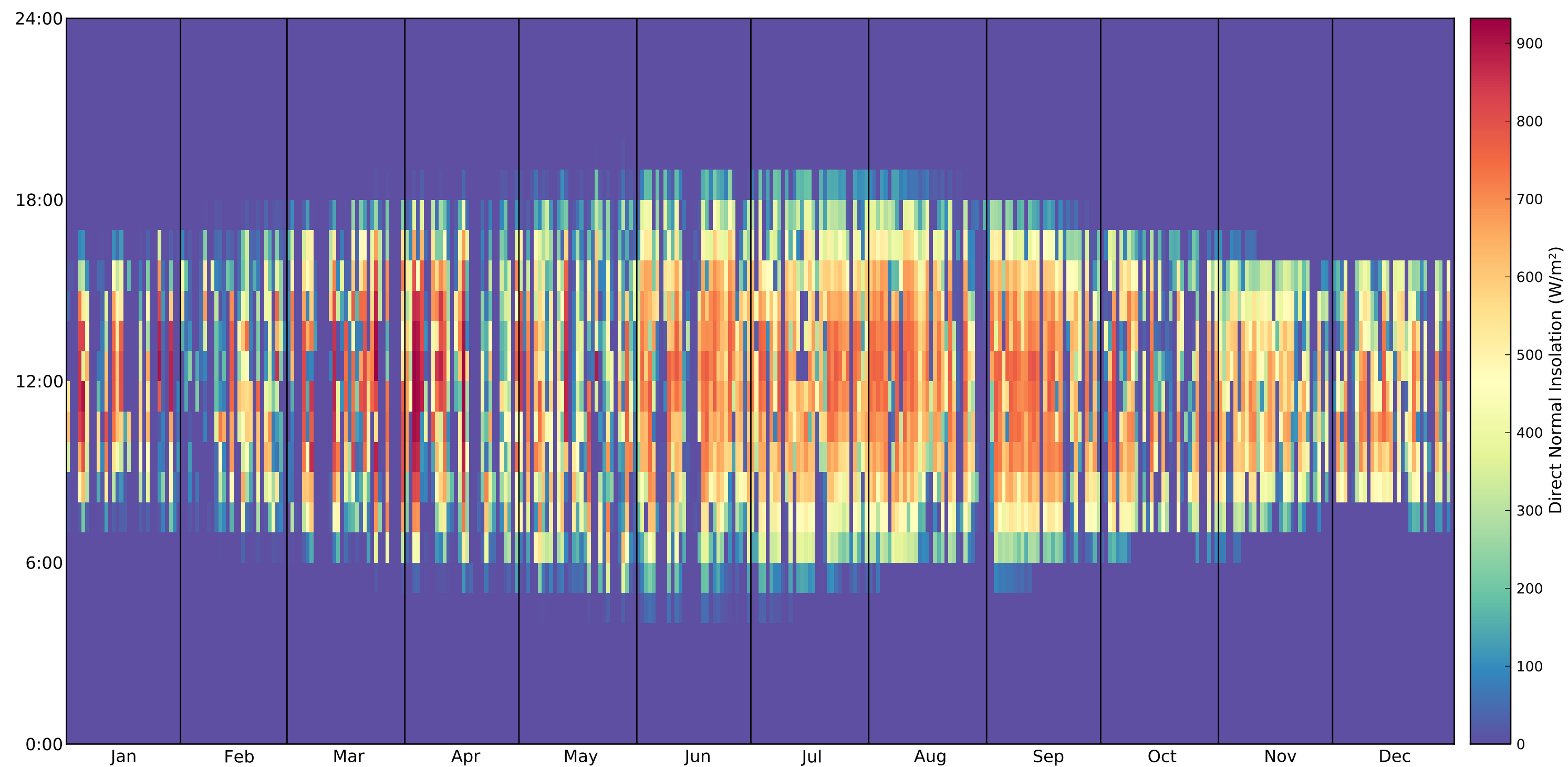


Figure 3: Direct normal insolation for a given day of year and hour of day.

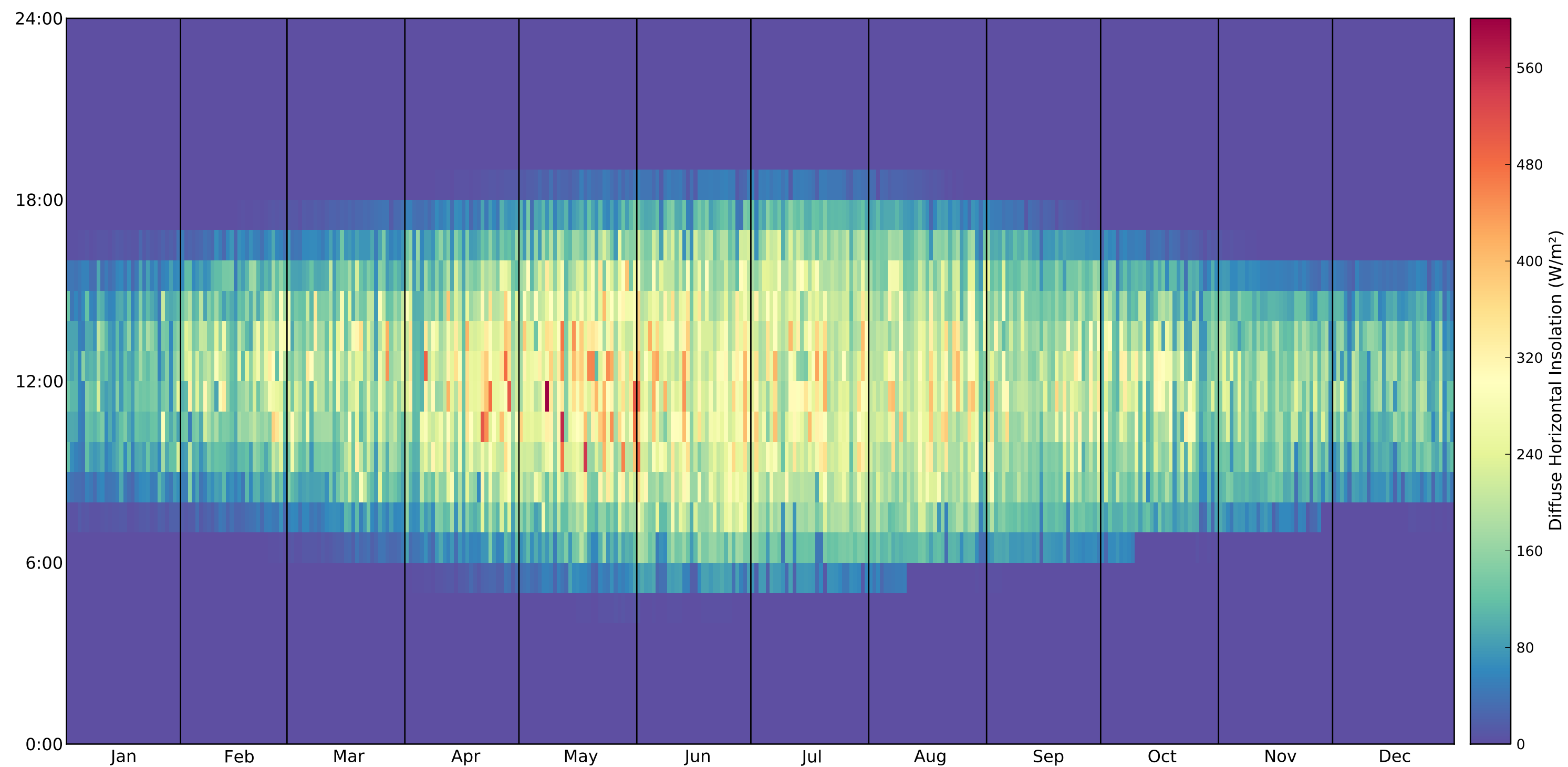


Figure 4: Diffuse horizontal insolation for a given day of year and hour of day.

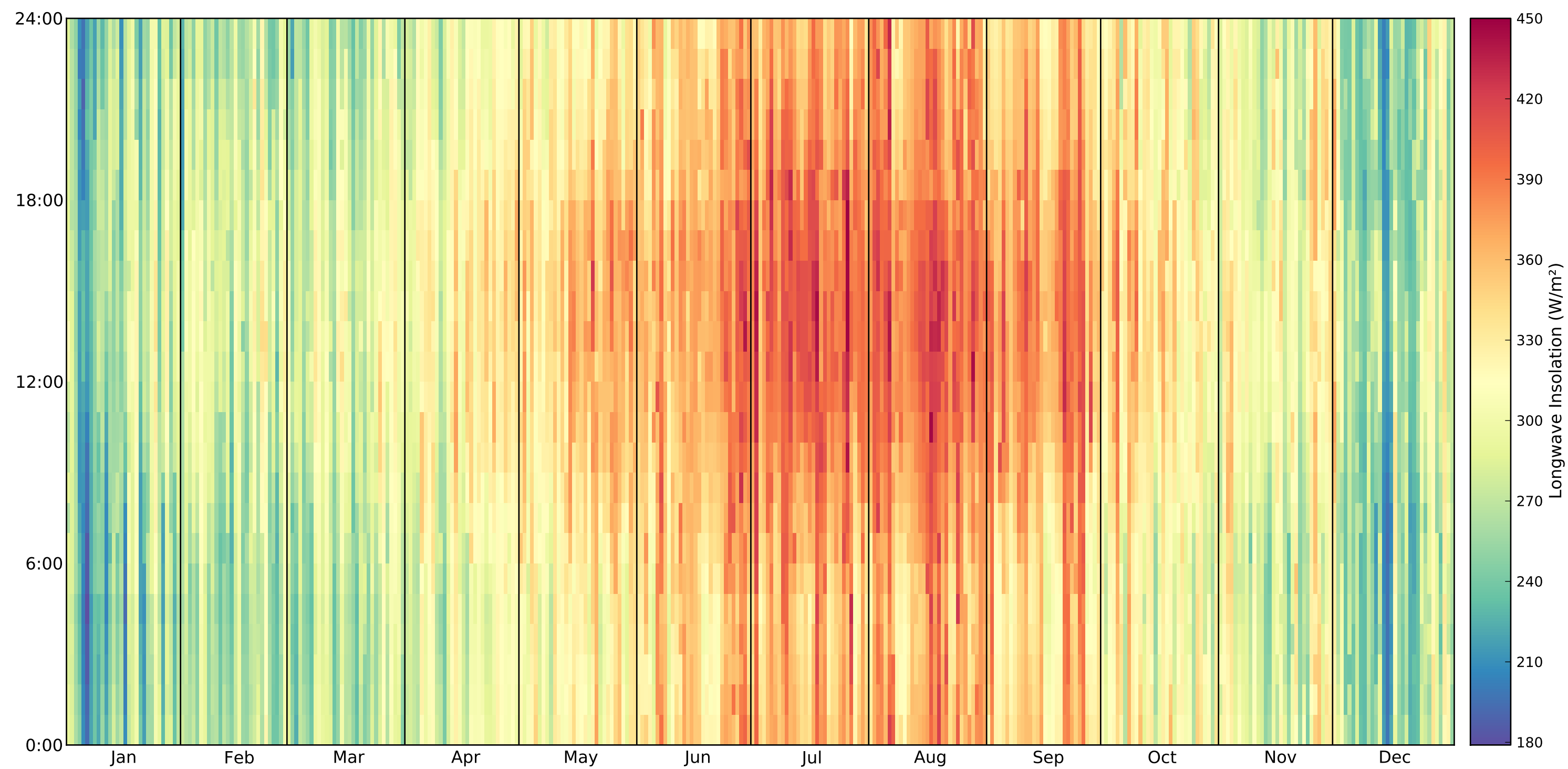


Figure 5: Longwave insolation for a given day of year and hour of day.

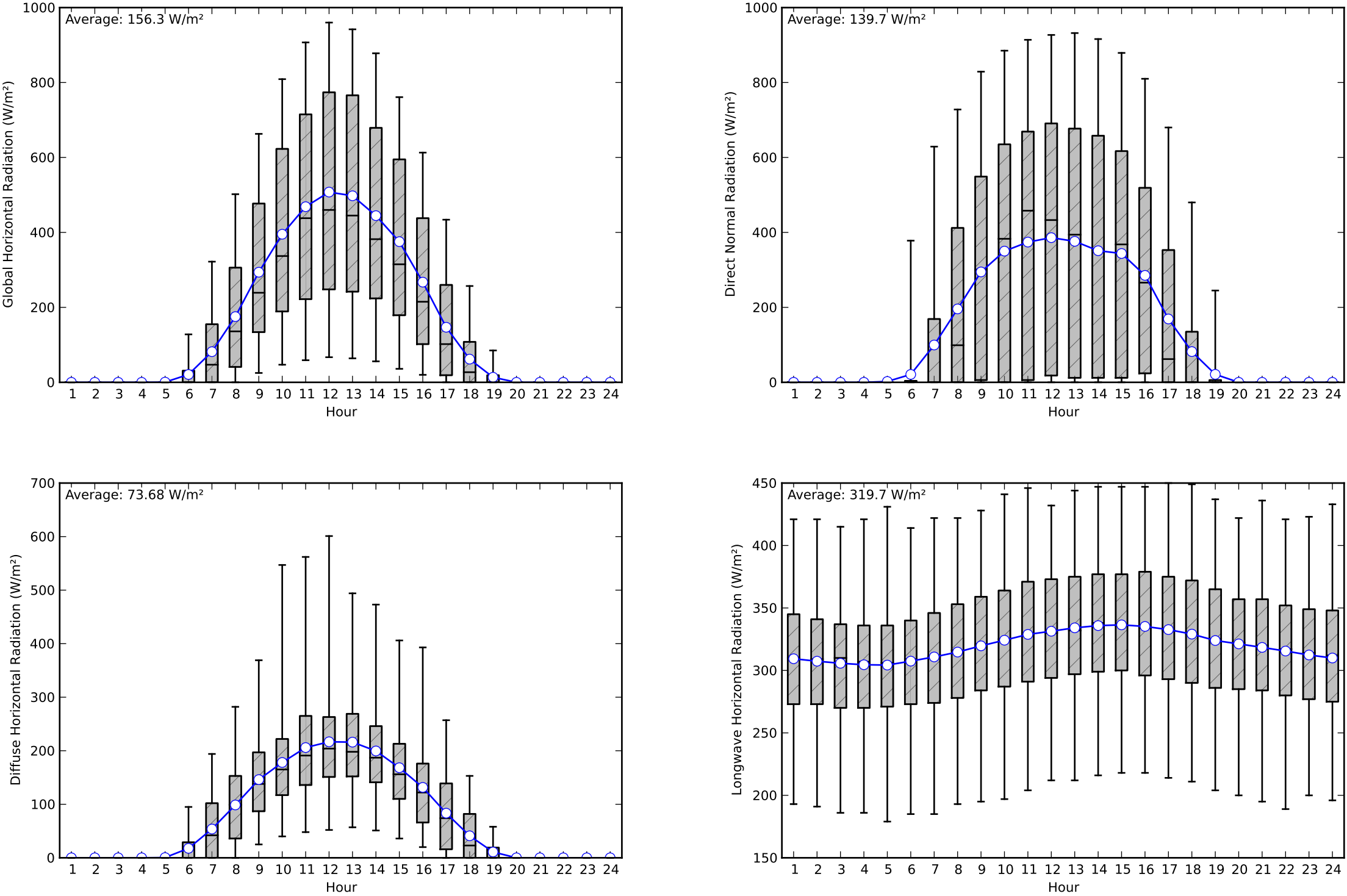


Figure 6: Hourly incoming direct normal, diffuse horizontal, global horizontal, and longwave radiation. Grey boxes indicate 25% and 75% percentile ranges while whiskers indicate minimum and maximum values.

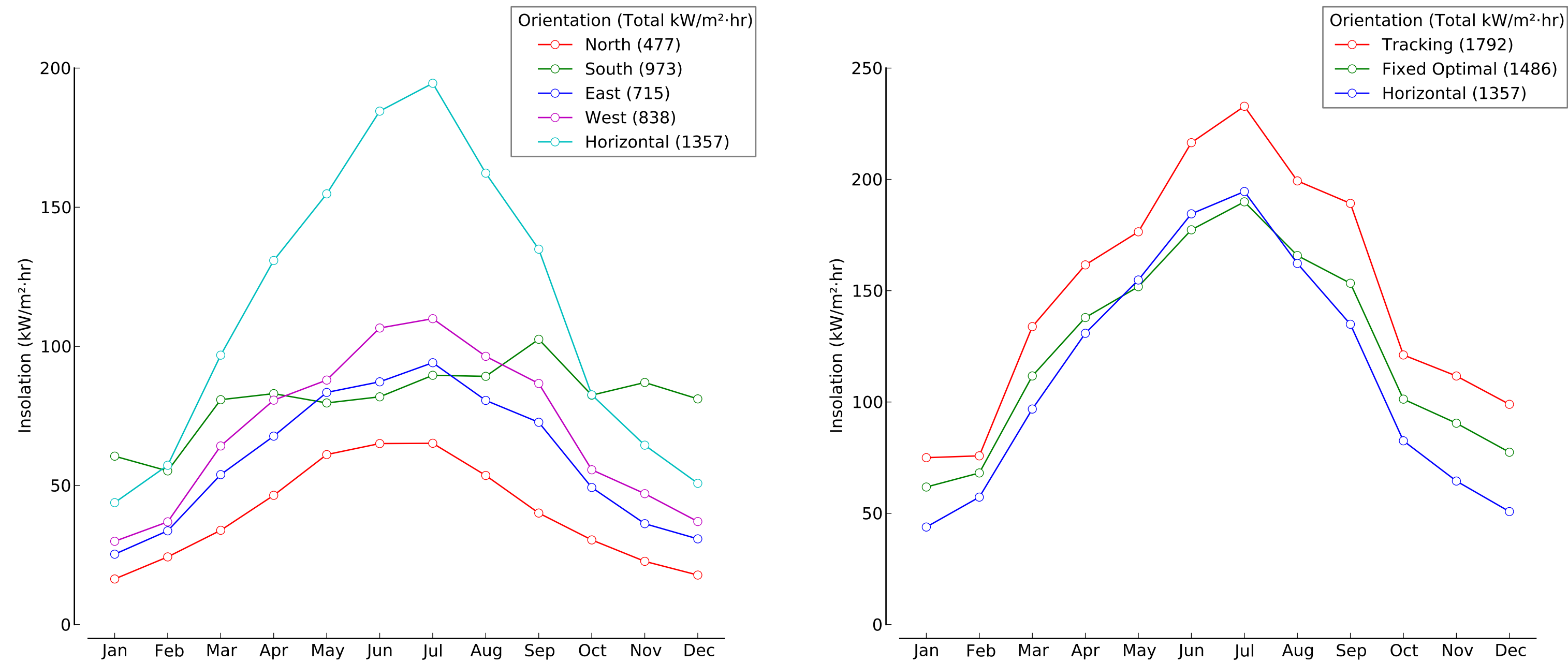


Figure 7: Total monthly solar insolation received on various inclined surfaces. Total annual amounts are indicated in the legend. The optimal fixed tilt from the horizontal at this location is 29°.

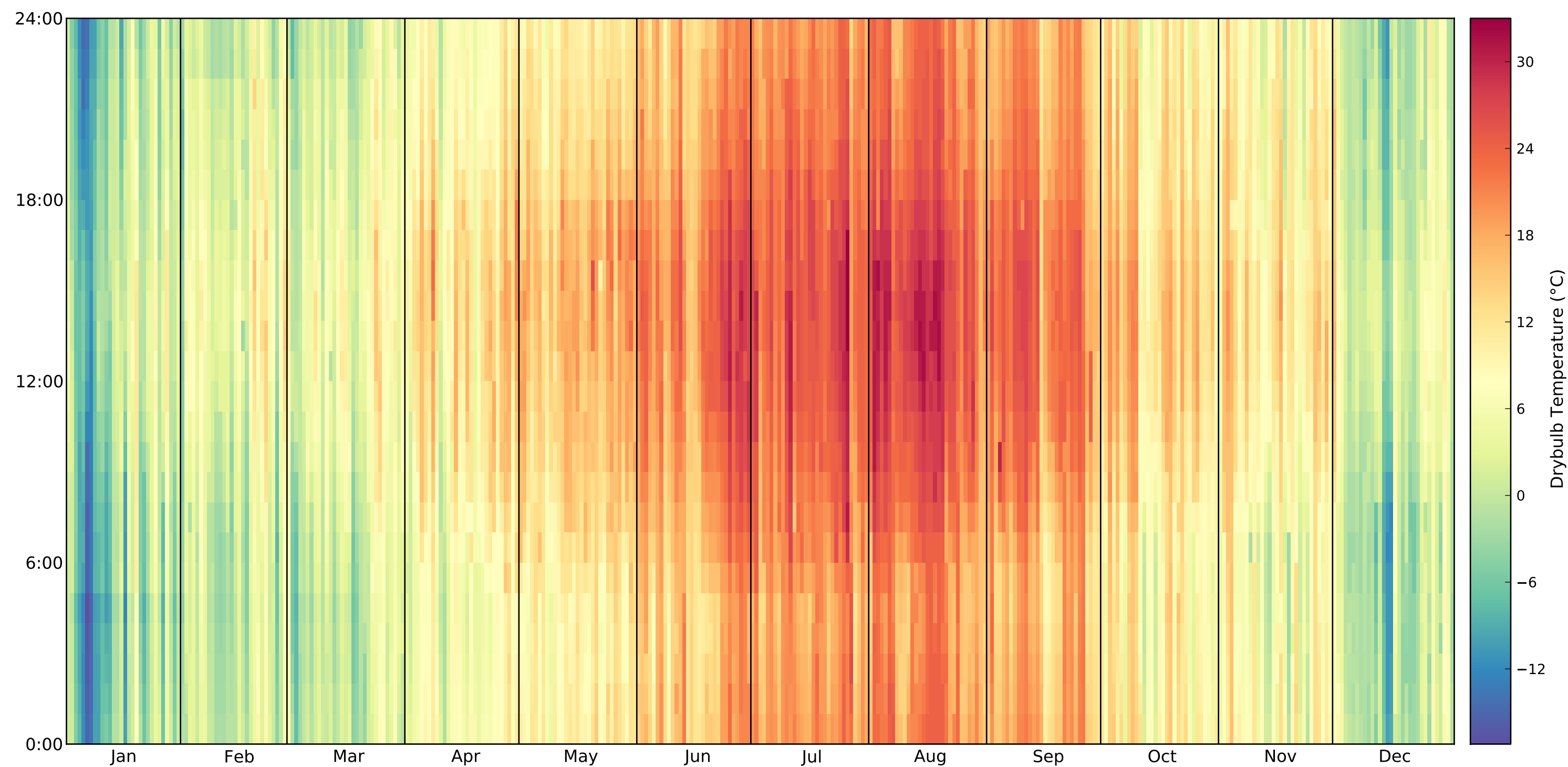


Figure 8: Drybulb temperature for a given day of year and hour of day.

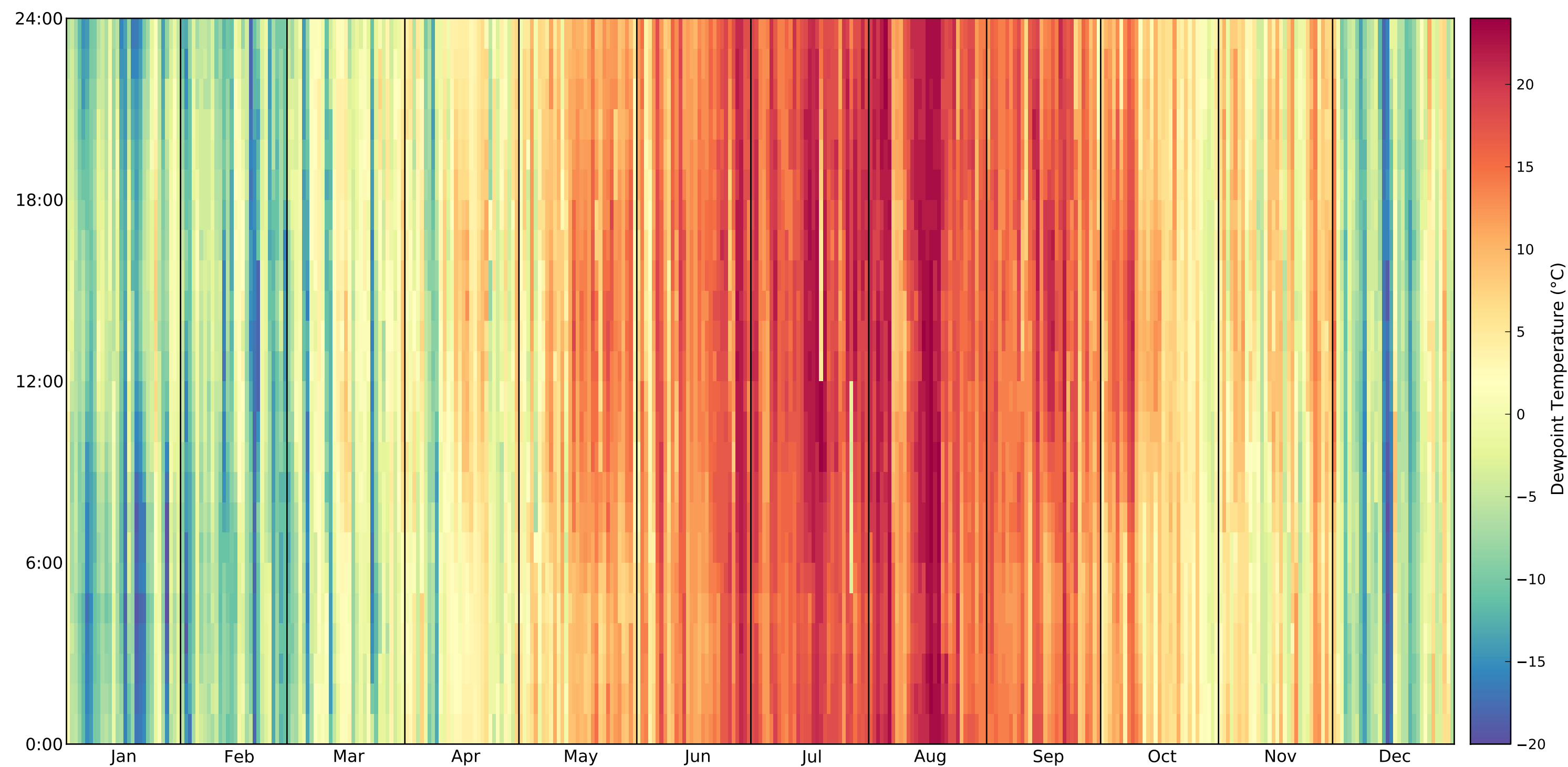


Figure 9: Dewpoint temperature for a given day of year and hour of day.

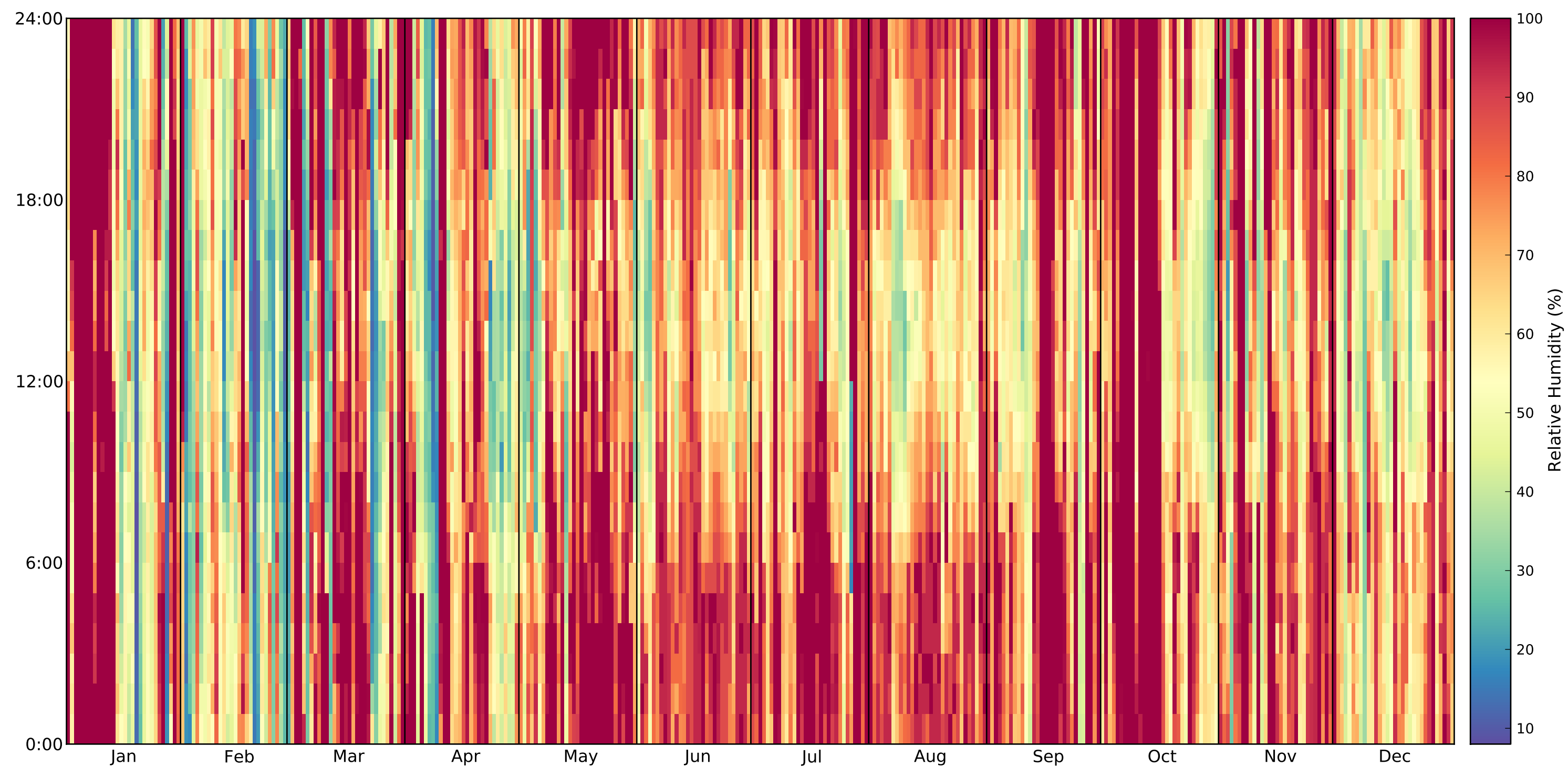


Figure 10: Relative humidity for a given day of year and hour of day.

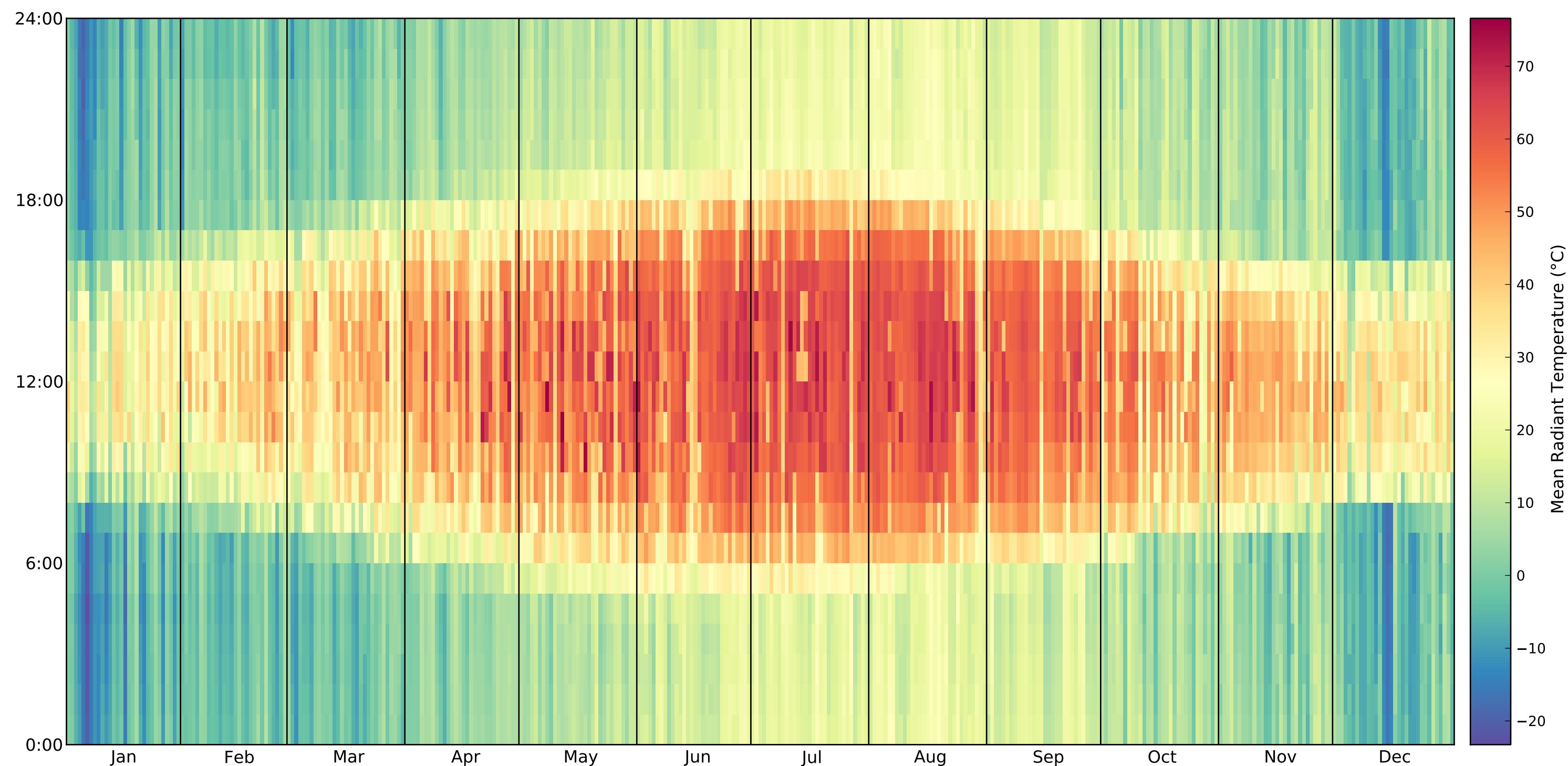


Figure 11: Mean radiant temperature for a given day of year and hour of day assuming fully-exposed to the effects of sun and wind.

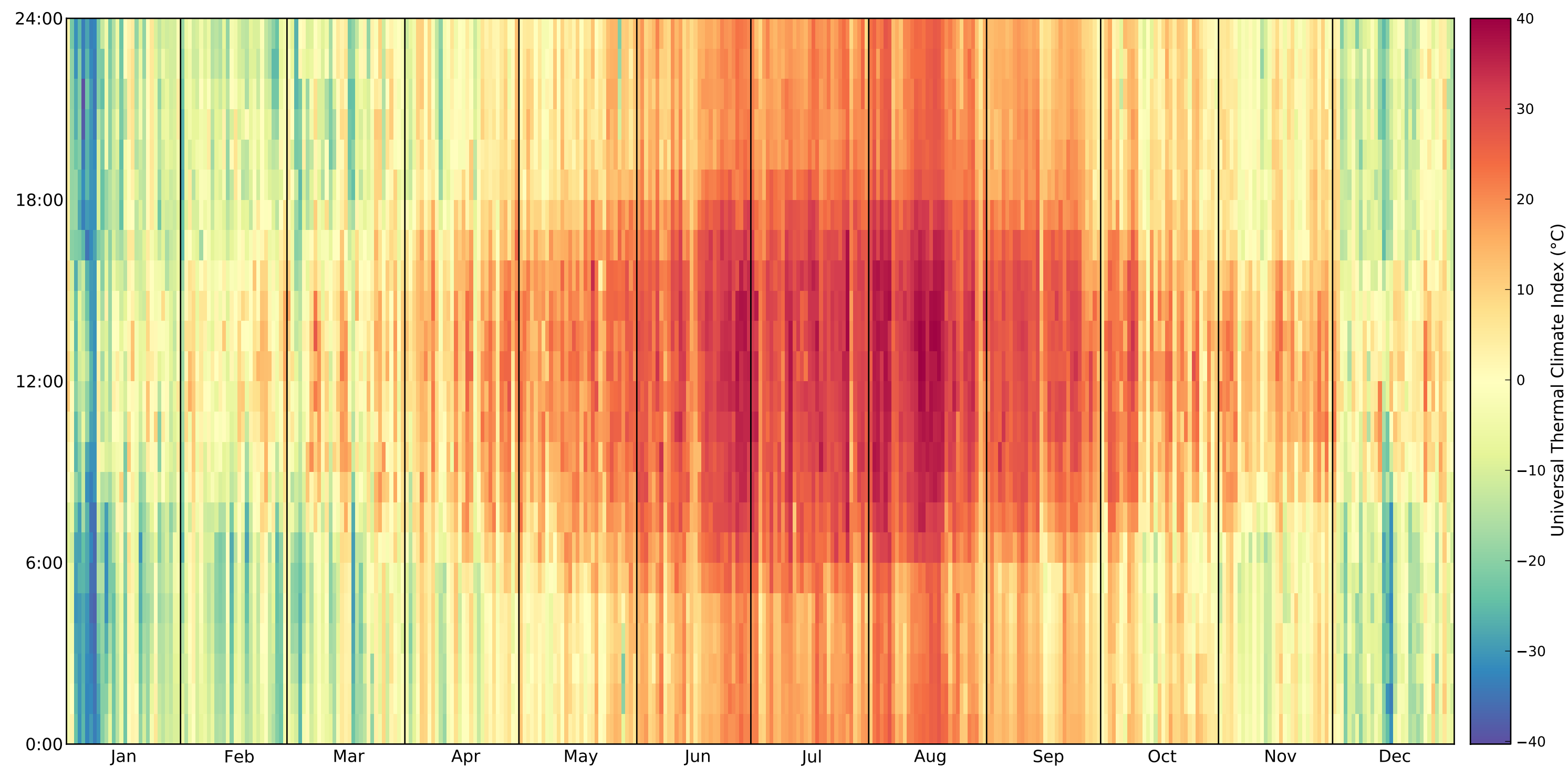


Figure 12: Universal Thermal Climate Index [UTCI](#) for a given day of year and hour of day assuming fully-exposed to the effects of sun and wind.

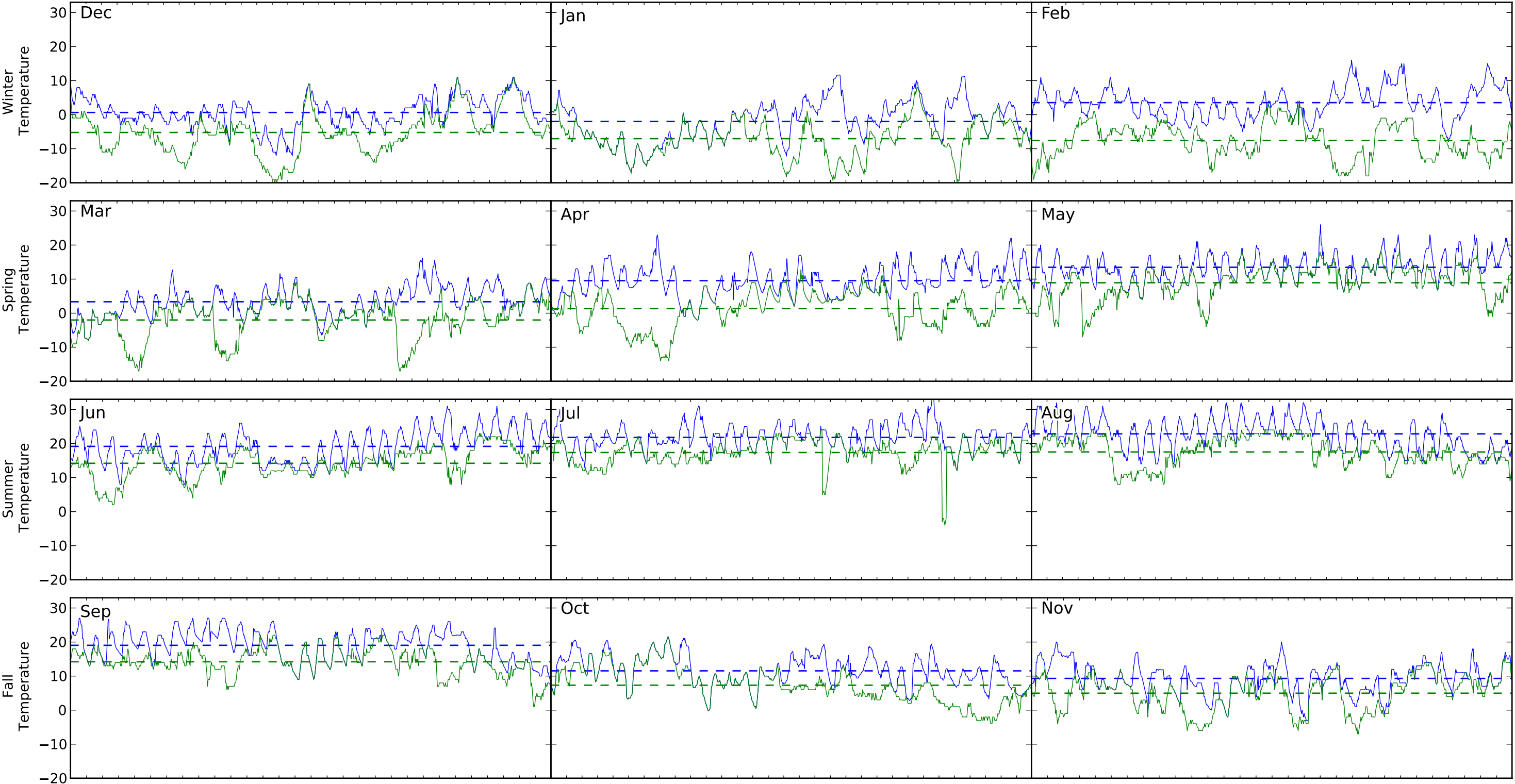


Figure 13: Full hourly time series of drybulb (blue) and dewpoint temperature (green).

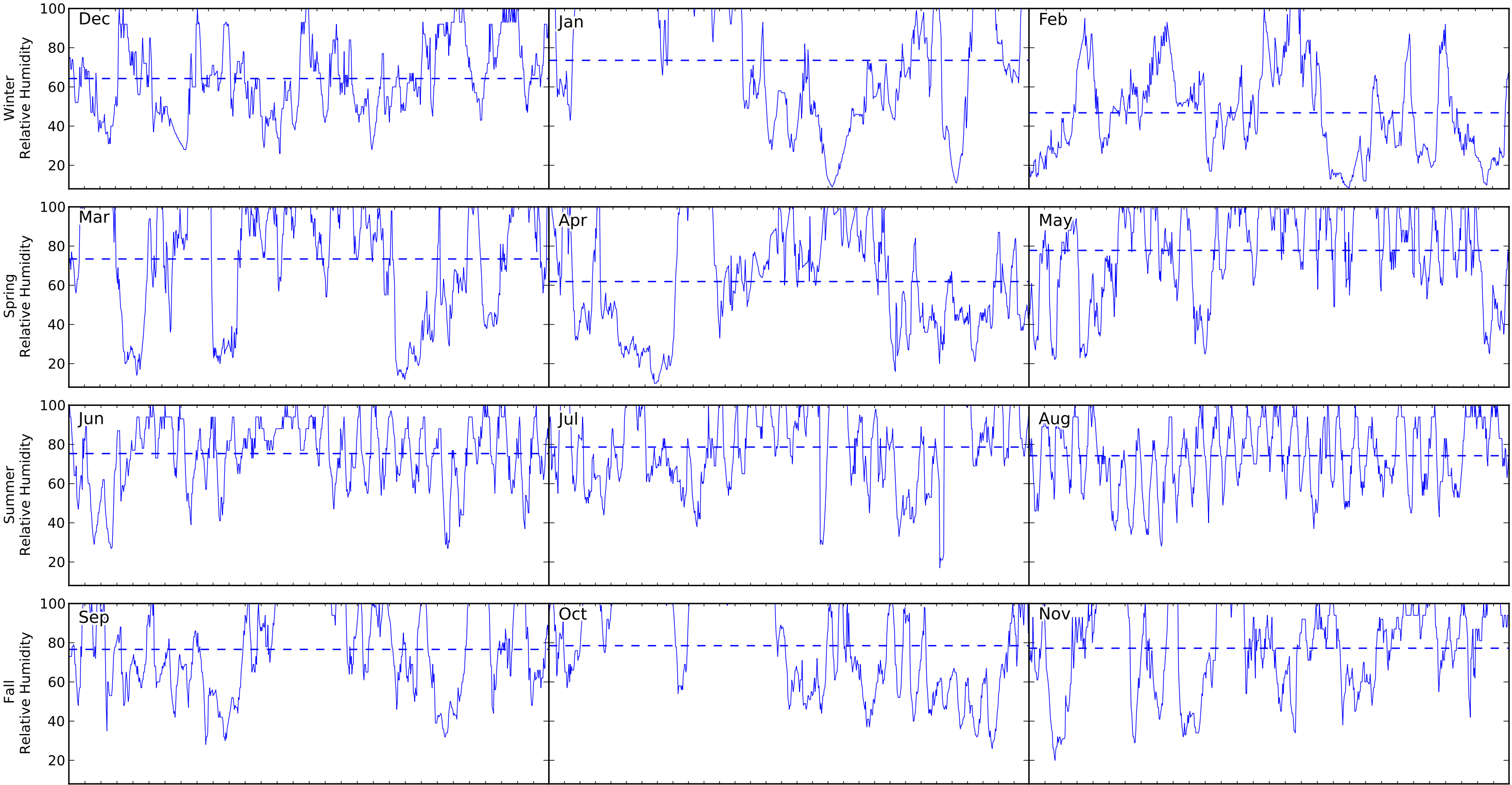


Figure 14: Full hourly time series of relative humidity.

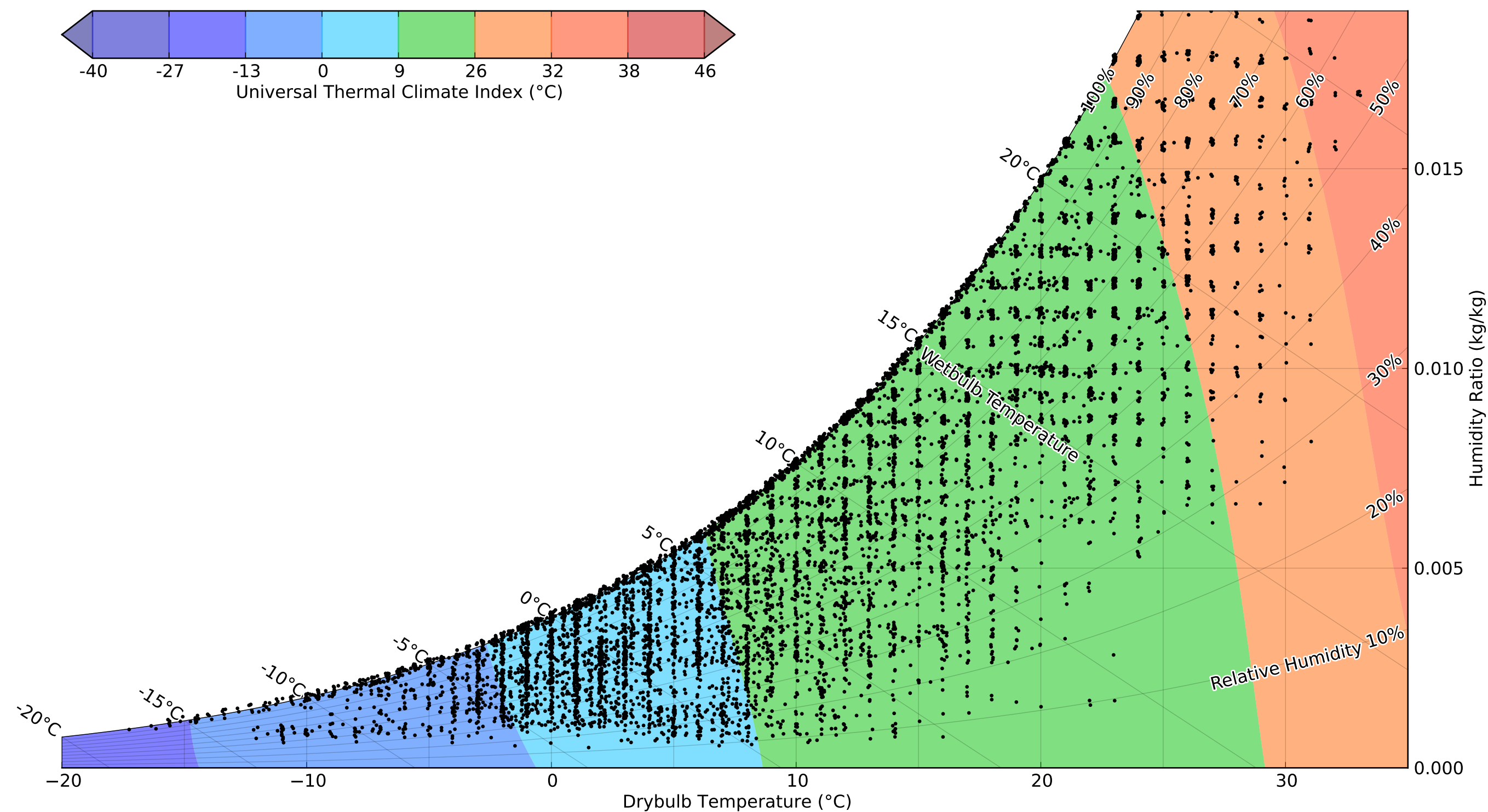
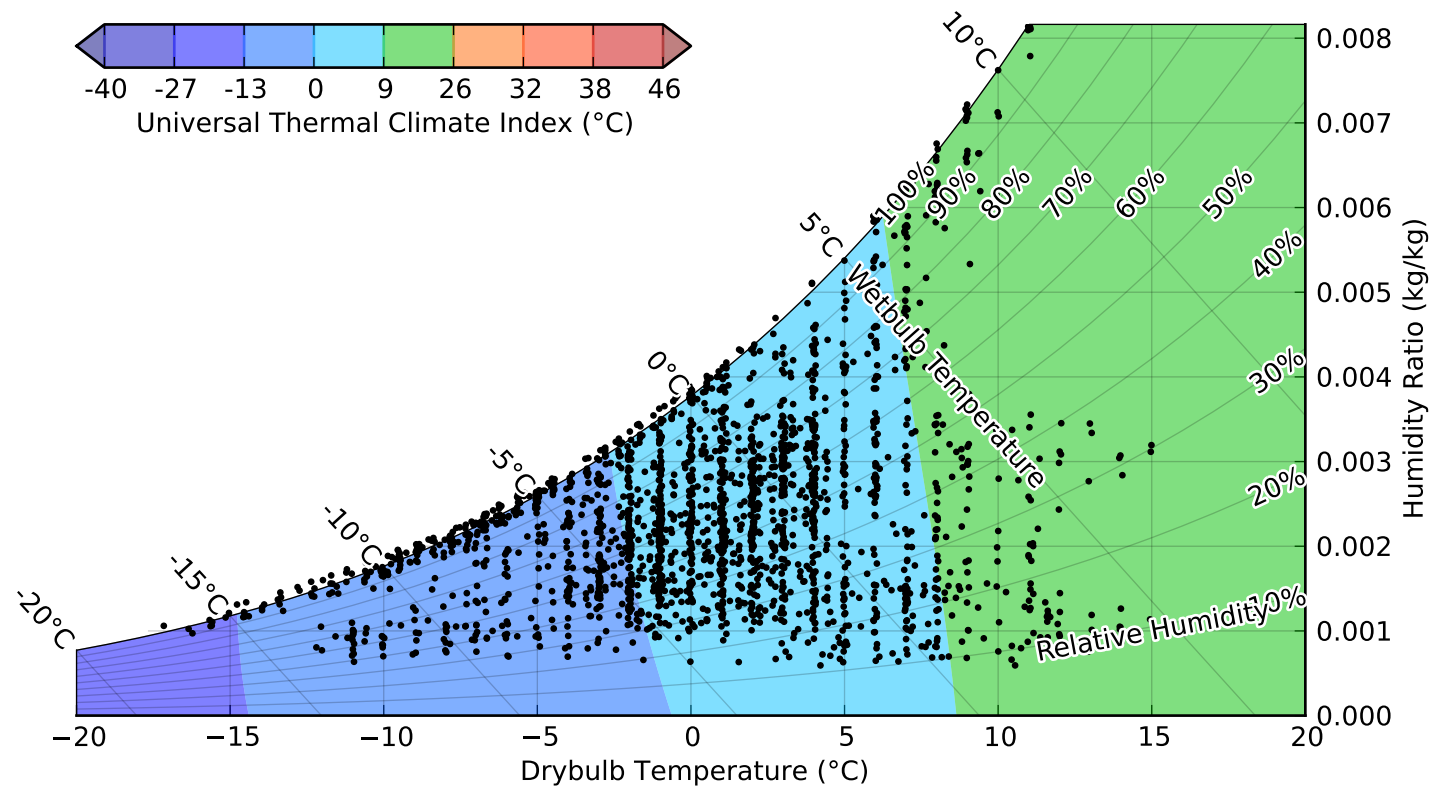
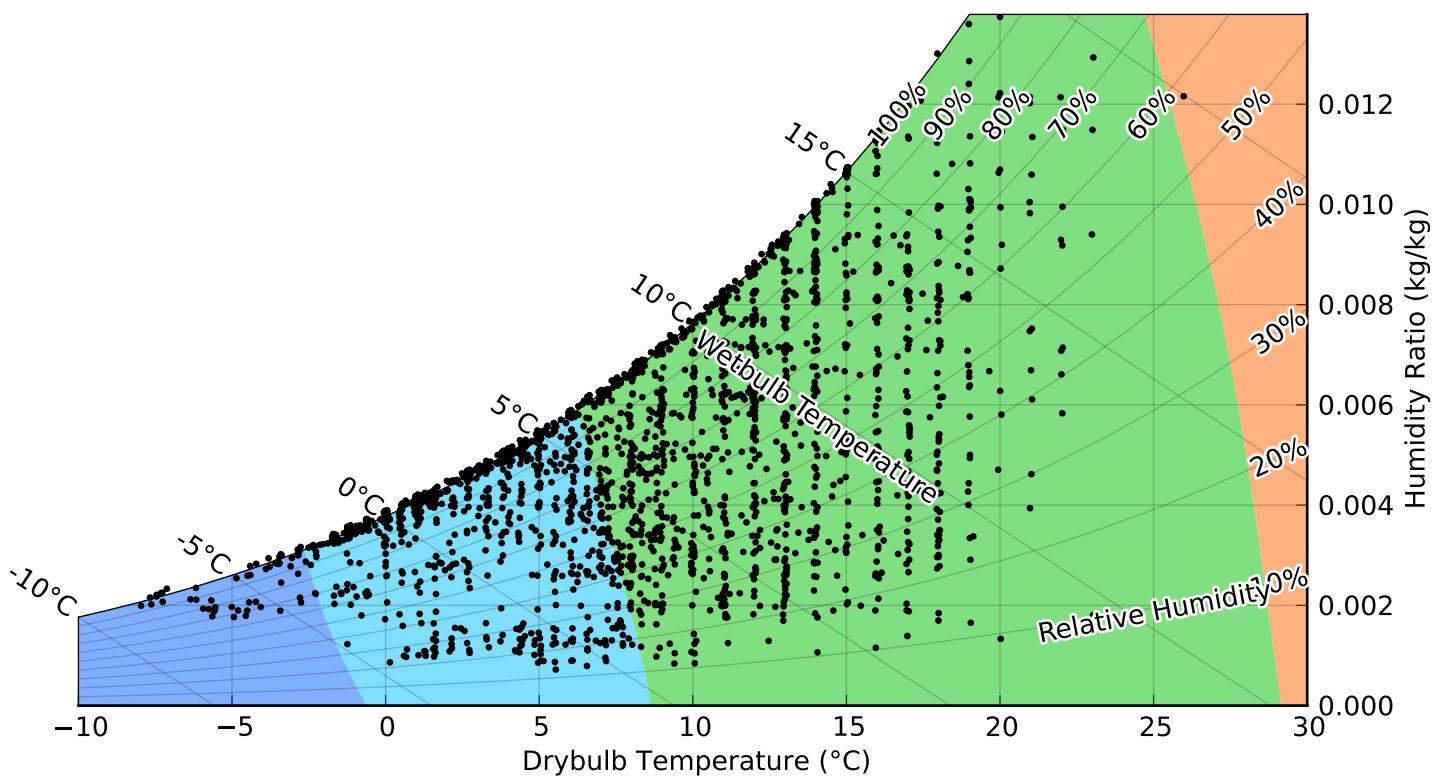


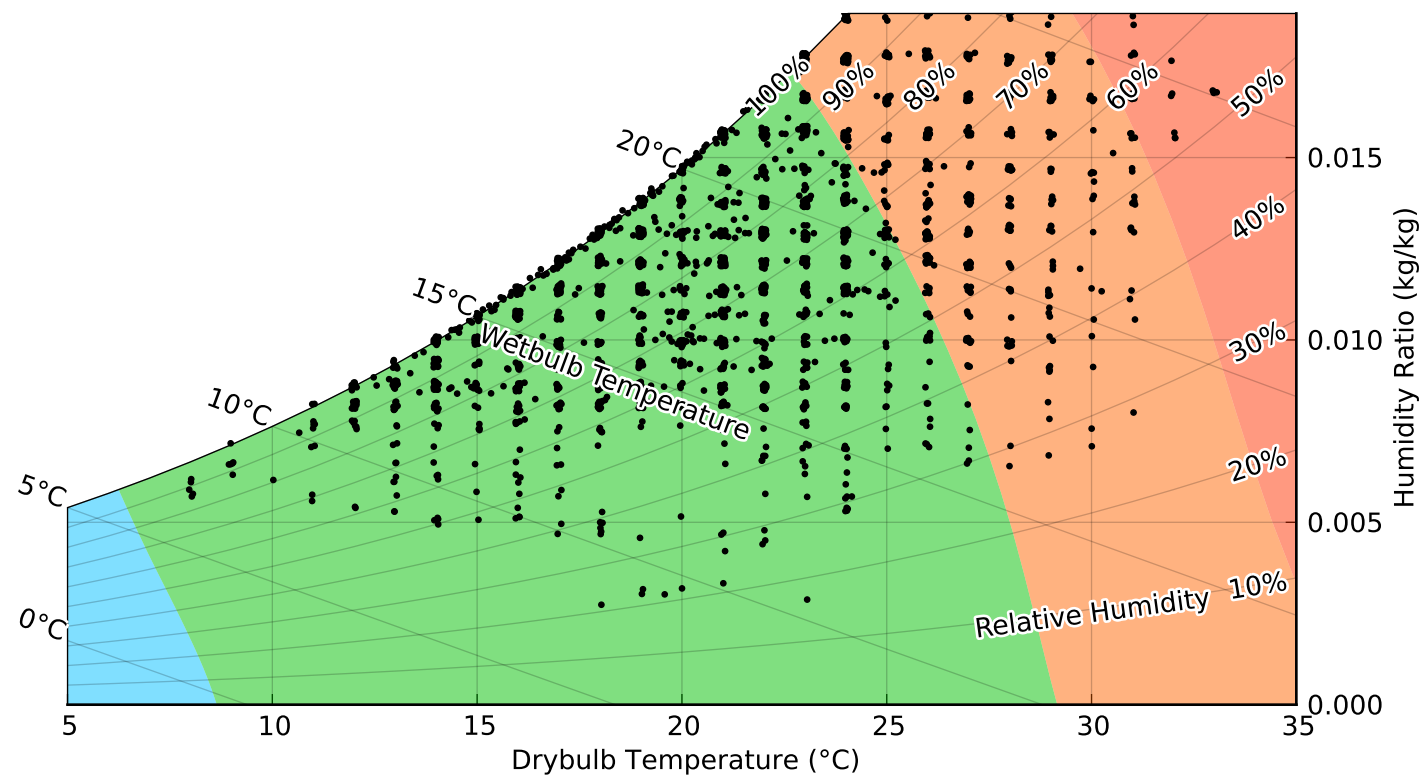
Figure 15: Pyschrometric plot of every hour in year. Contours of **UTCI** show the various thermal comfort regimes



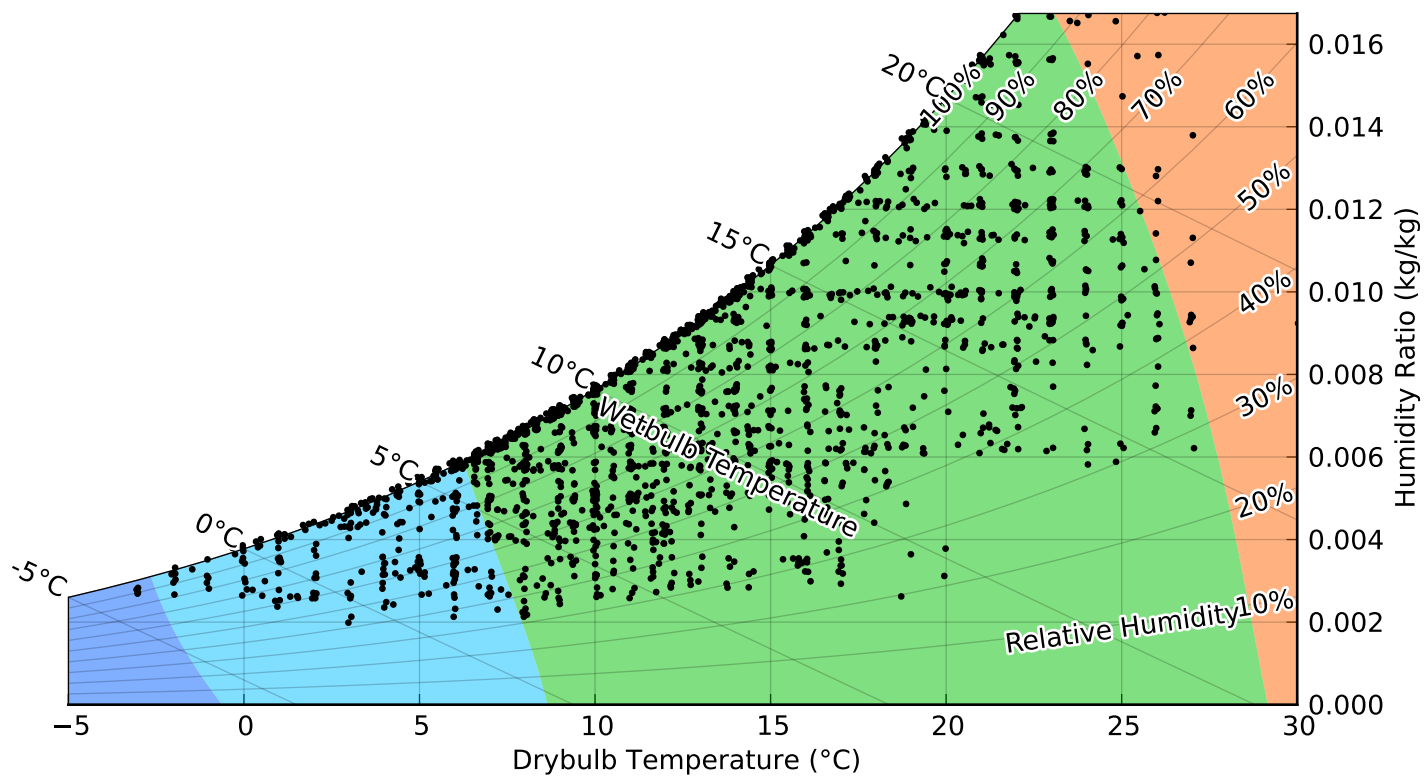
(a) Winter



(b) Spring



(c) Summer



(d) Autumn

Figure 16: Psychrometric plot of every hour in each season.

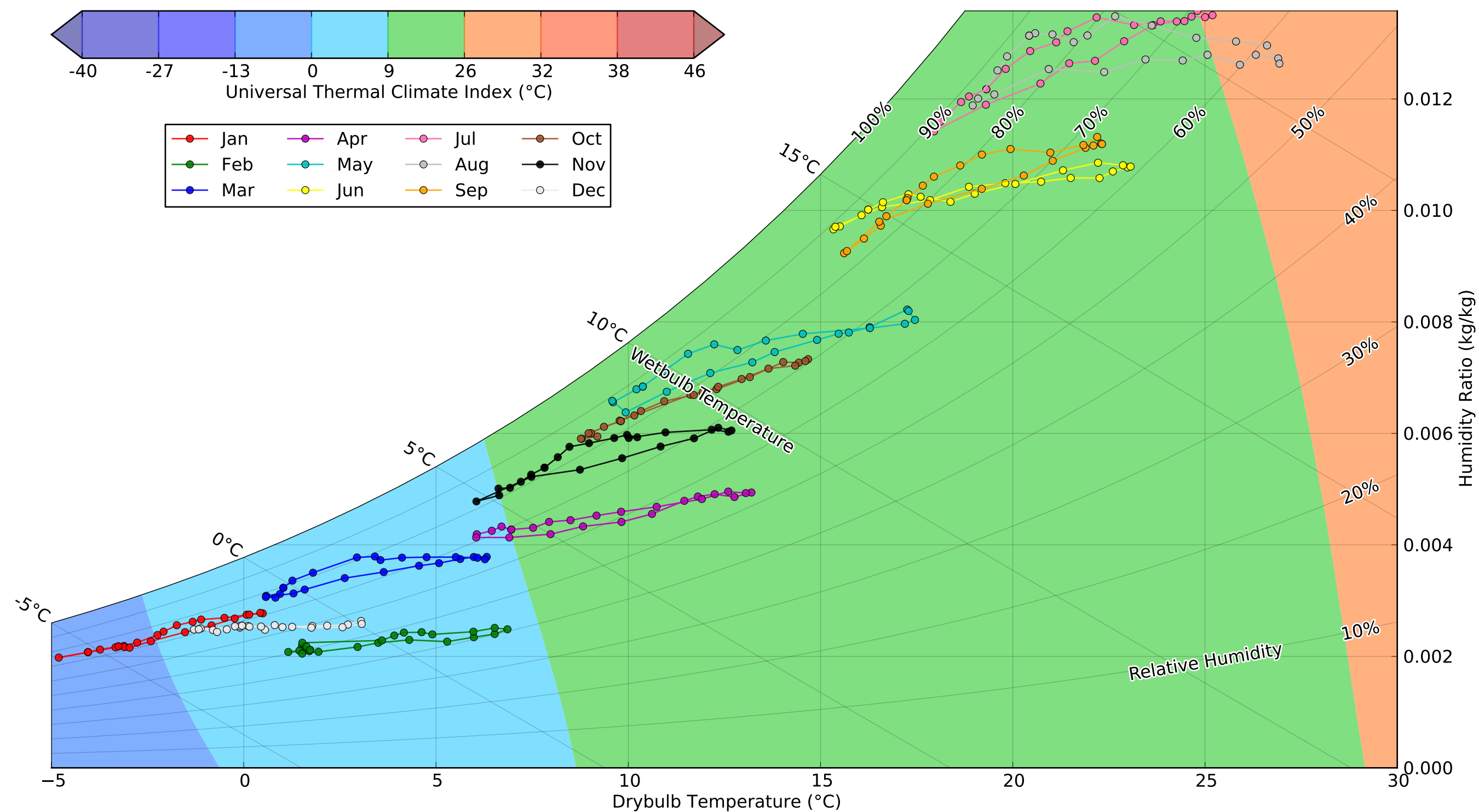
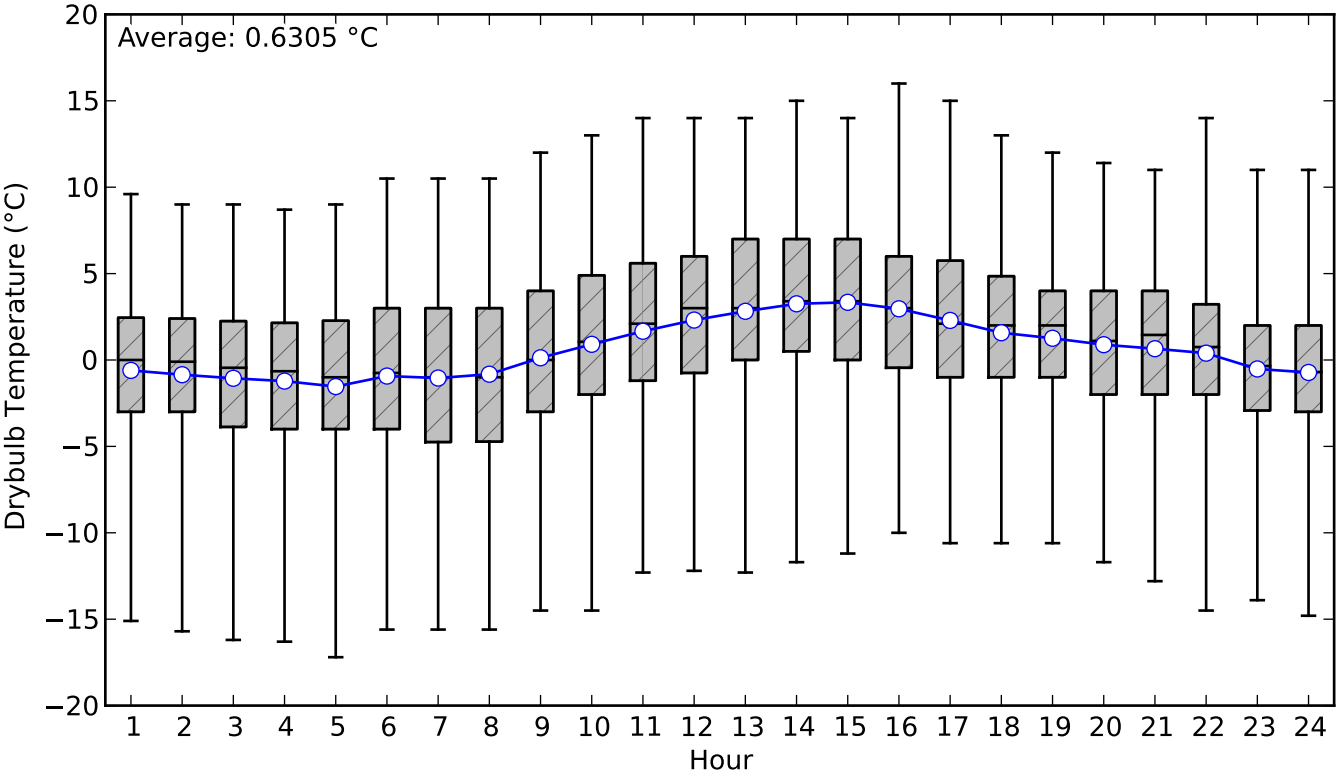
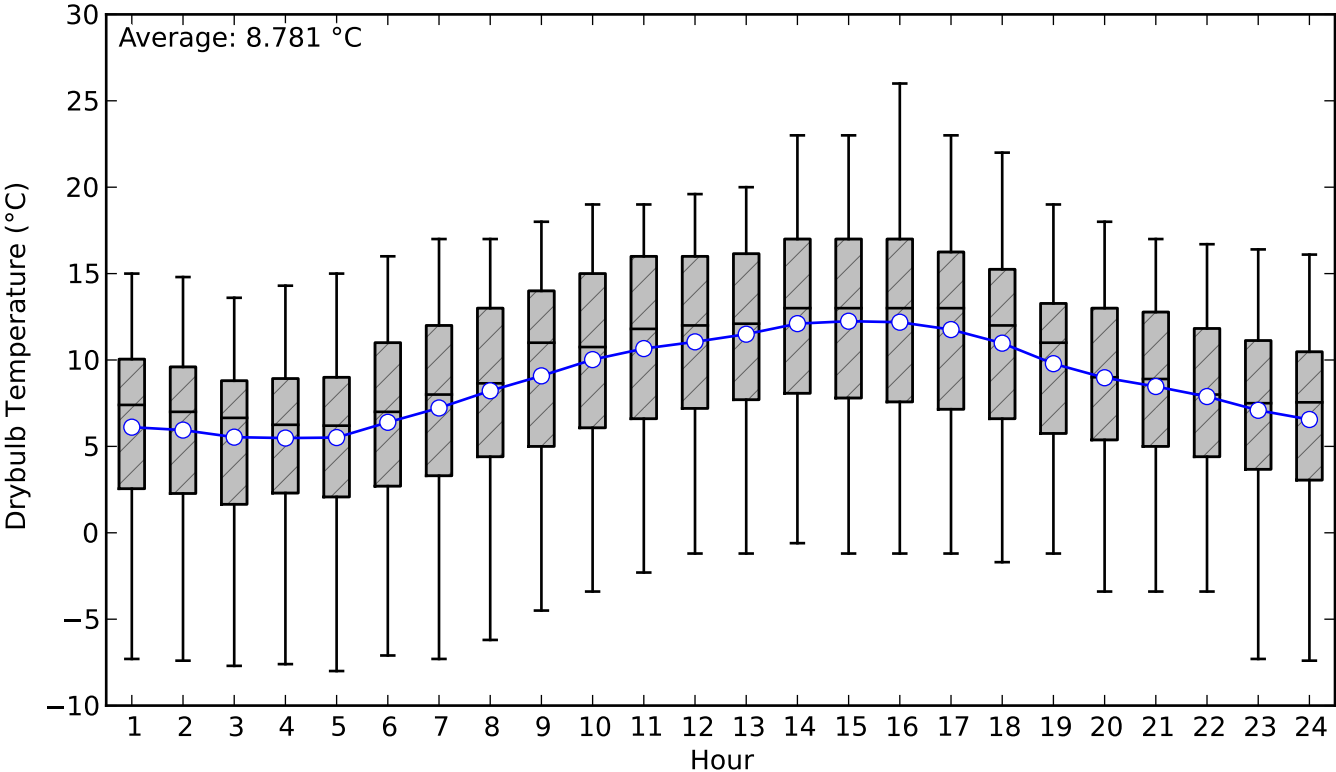


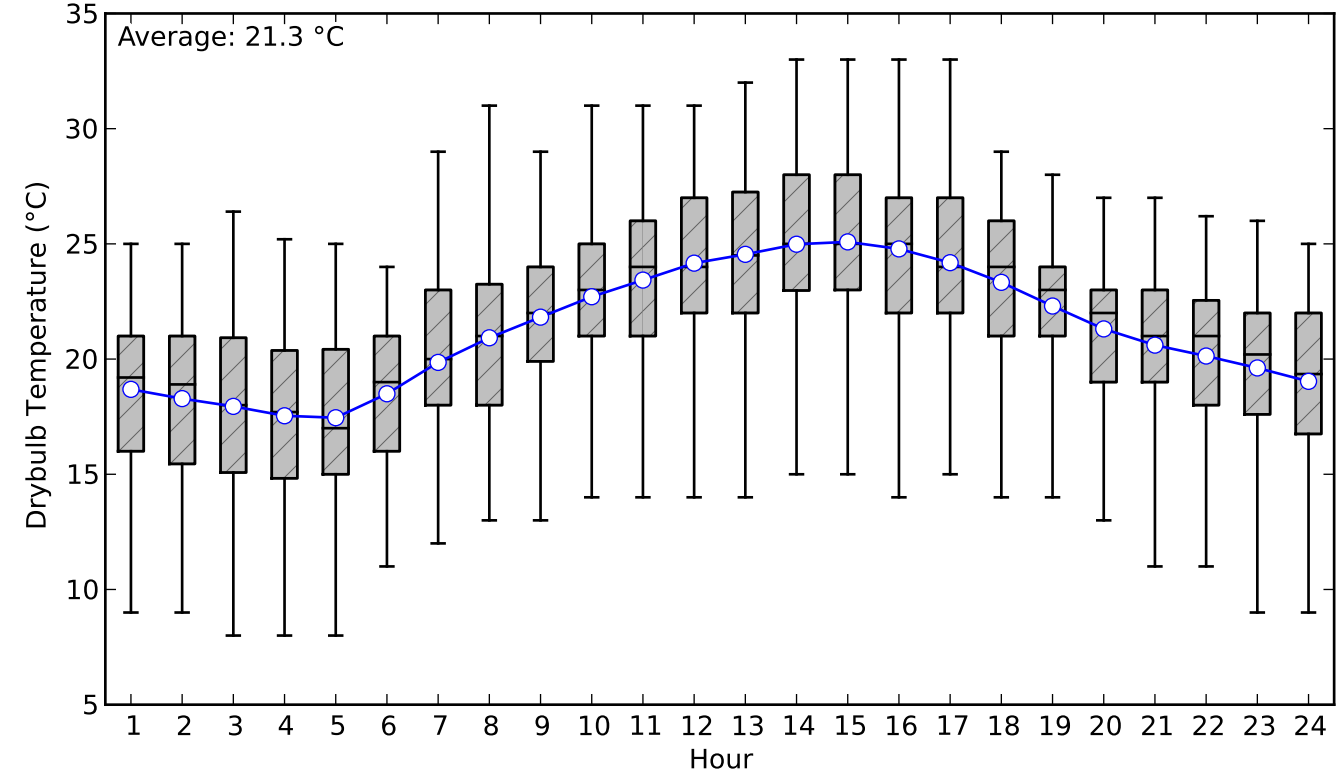
Figure 17: Hourly (24h) psychrometric paths for each month. Contours of UTCI show the various thermal comfort regimes



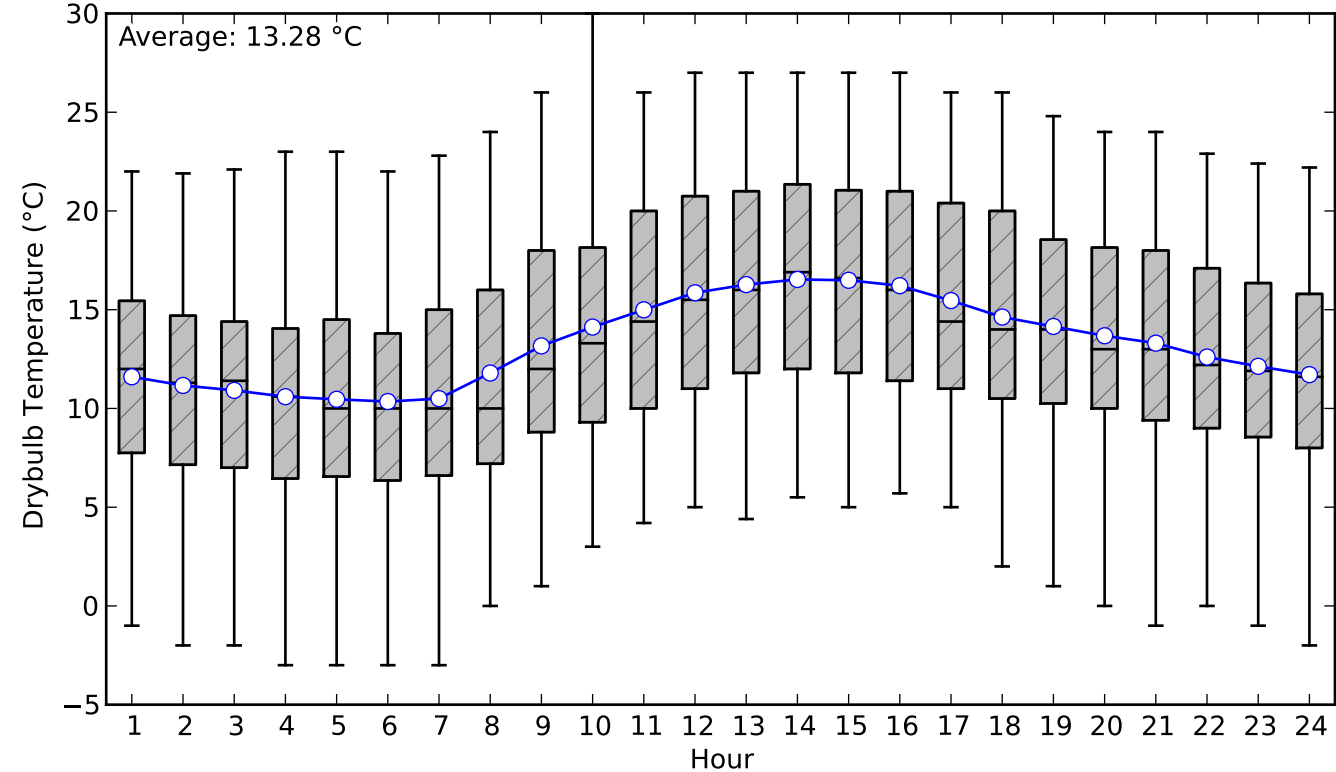
(a) Winter



(b) Spring

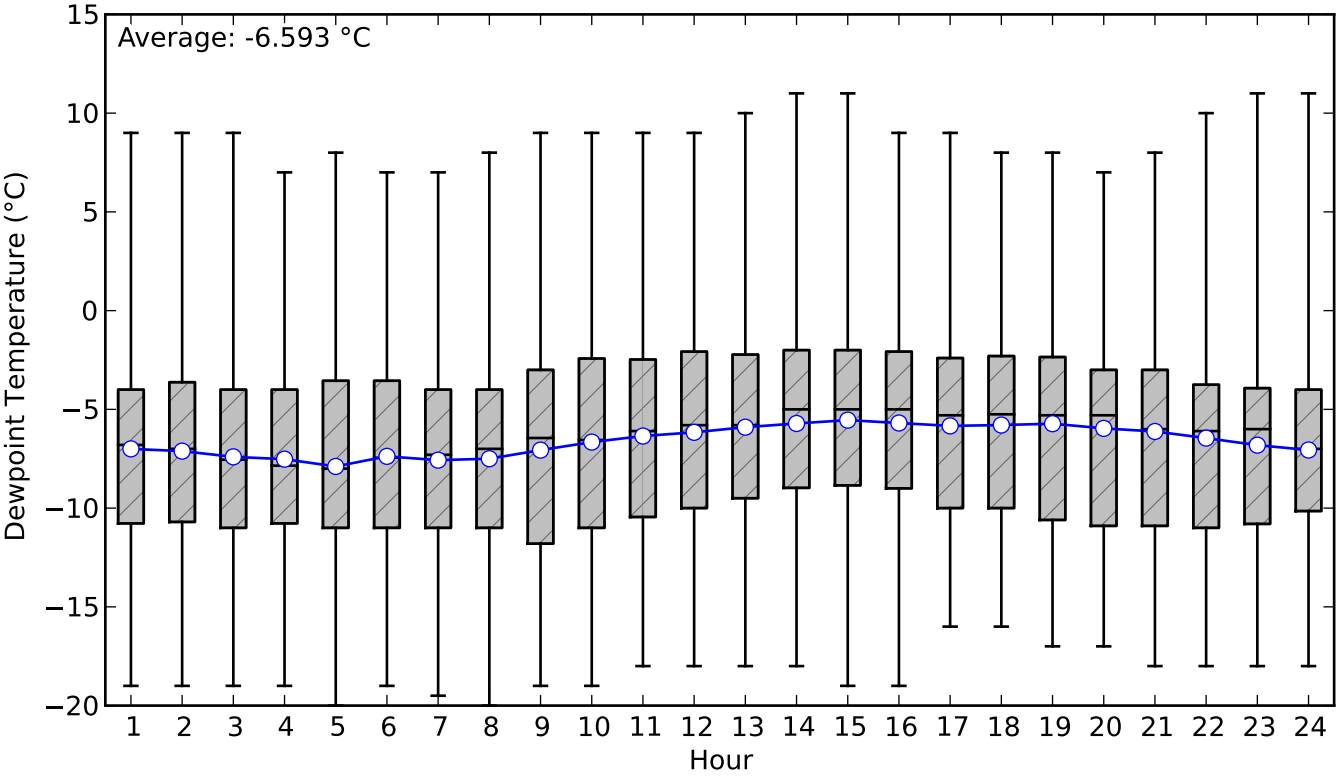


(c) Summer

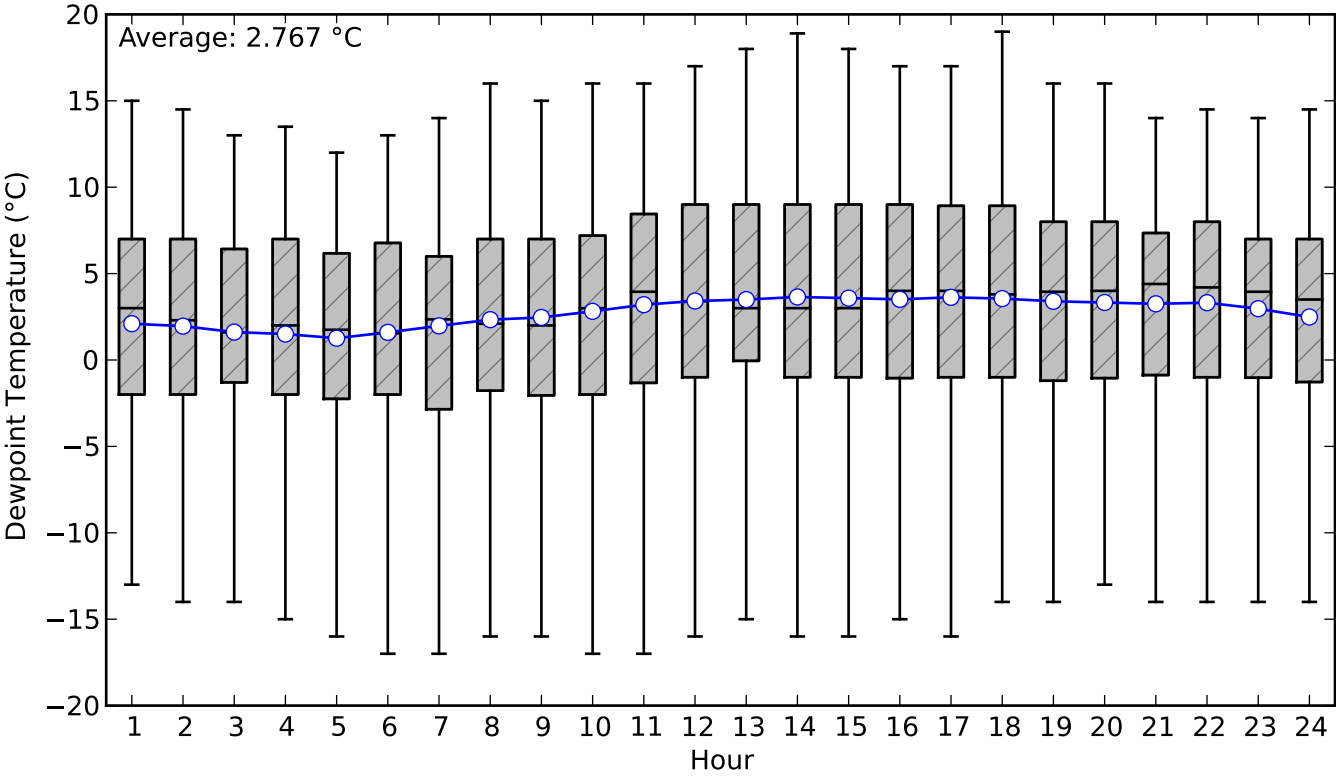


(d) Autumn

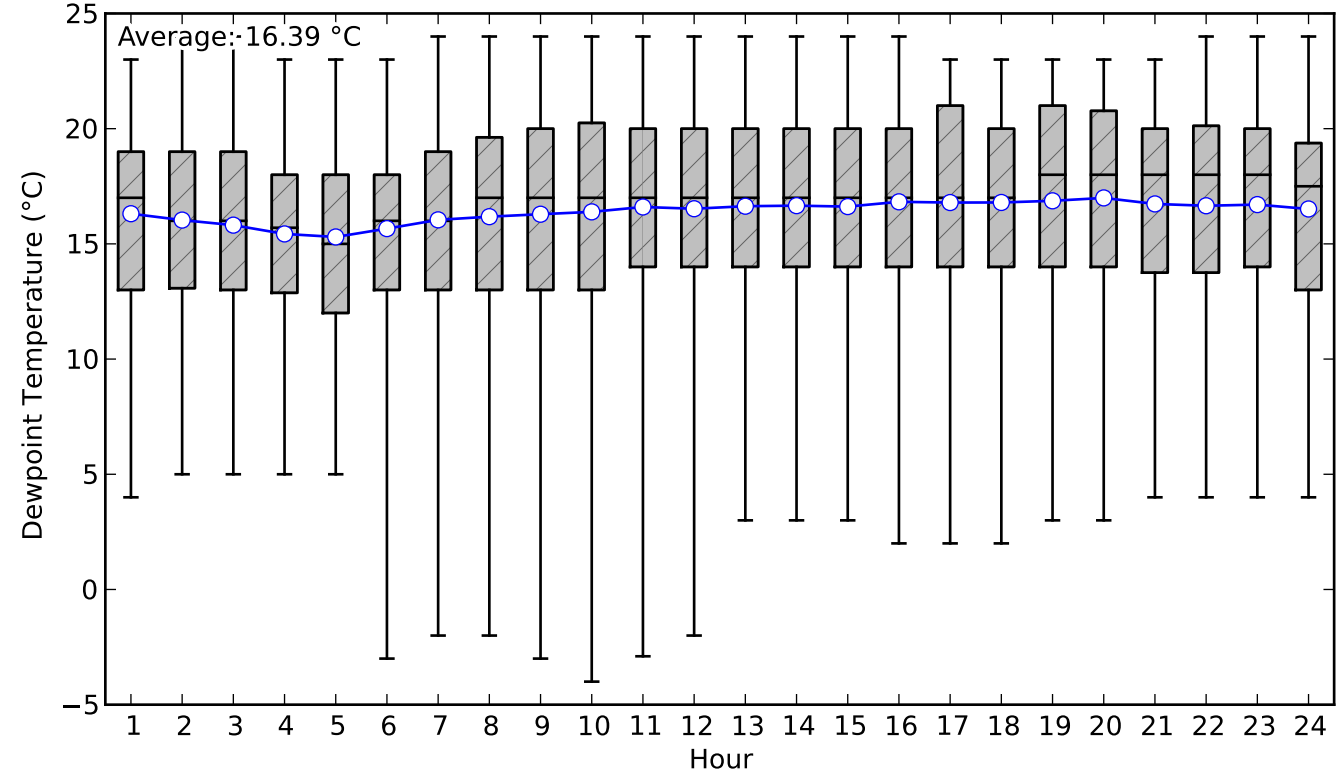
Figure 18: Hourly drybulb temperature for each season. Grey boxes indicate 25% and 75% percentile ranges while whiskers indicate minimum and maximum values. The blue line indicates the mean value.



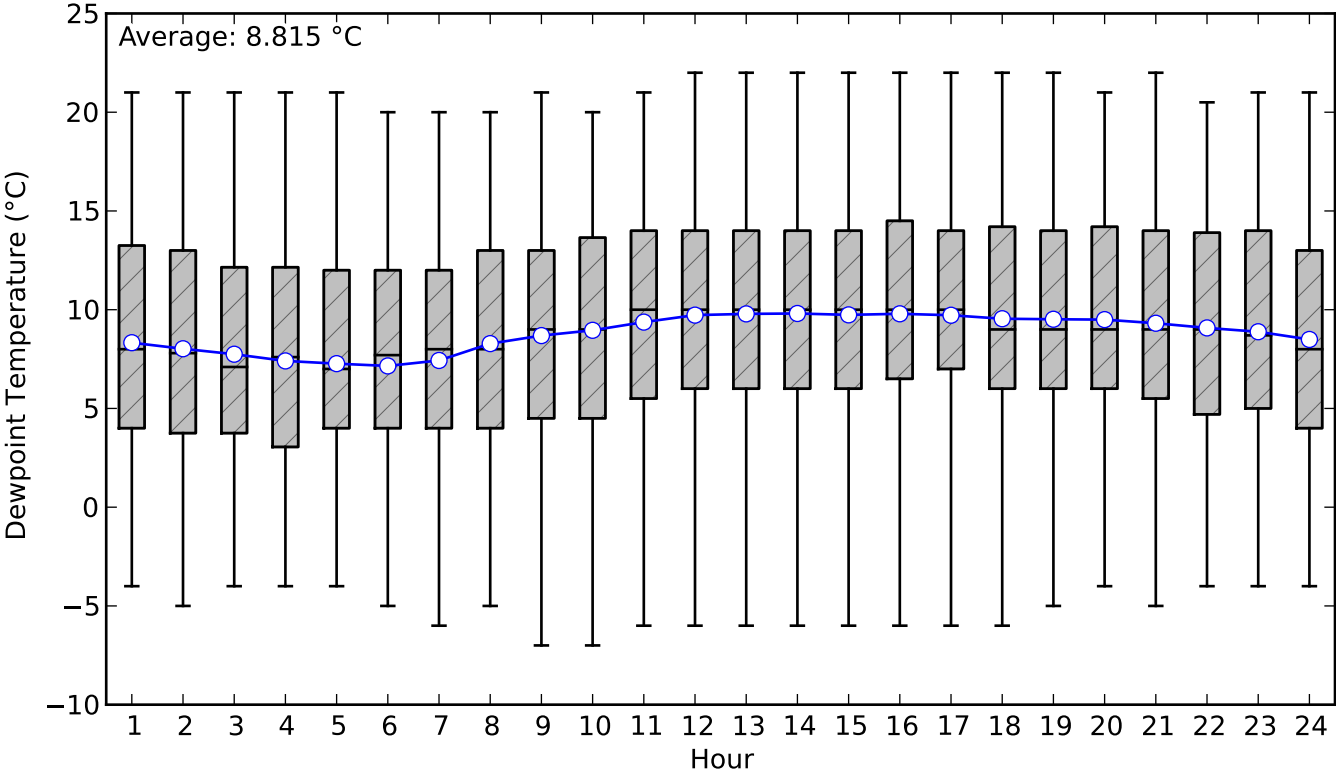
(a) Winter



(b) Spring

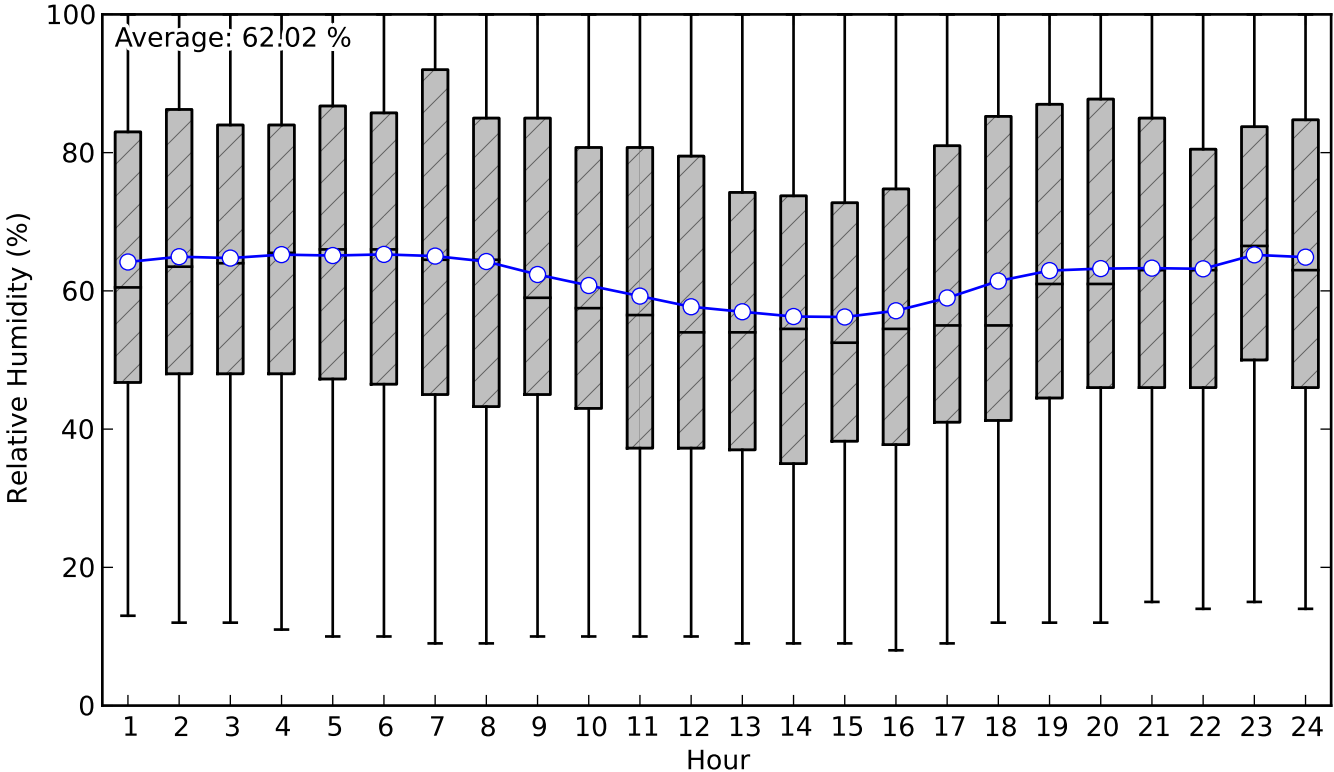


(c) Summer

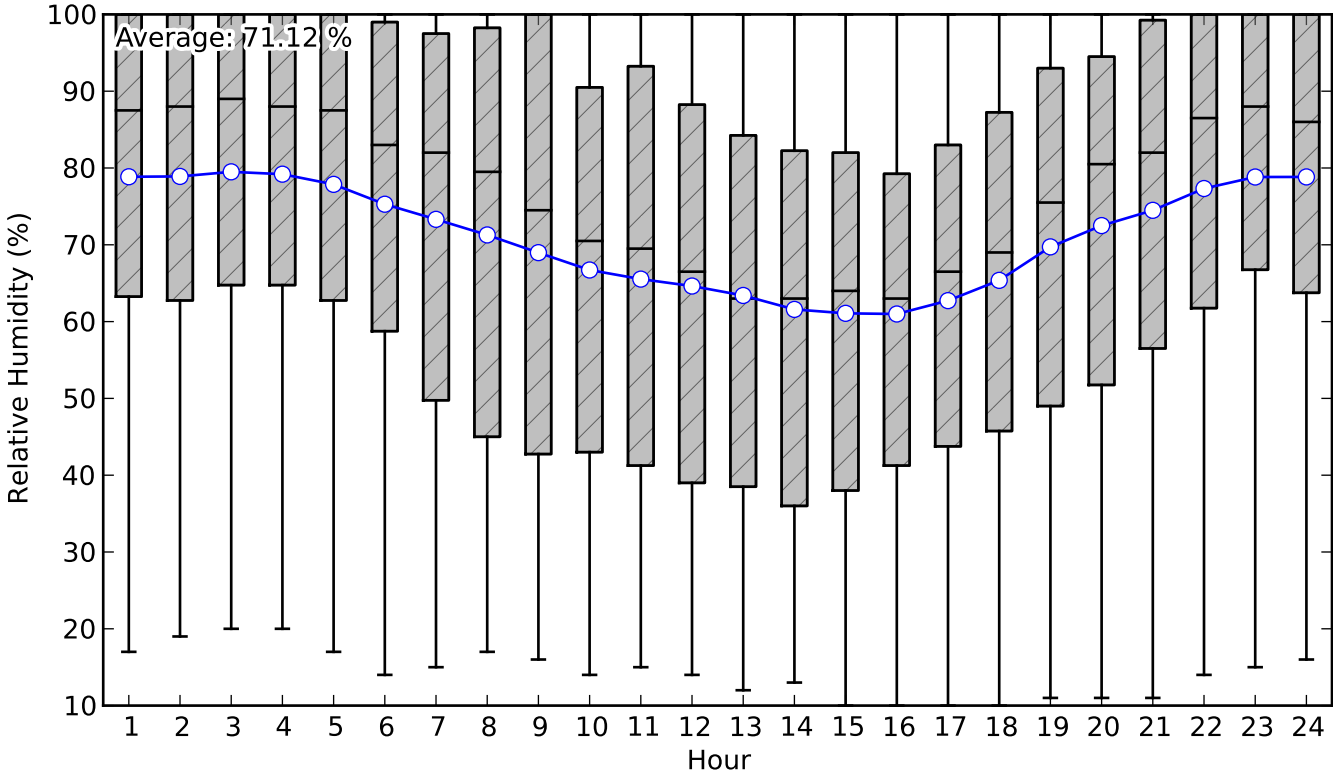


(d) Autumn

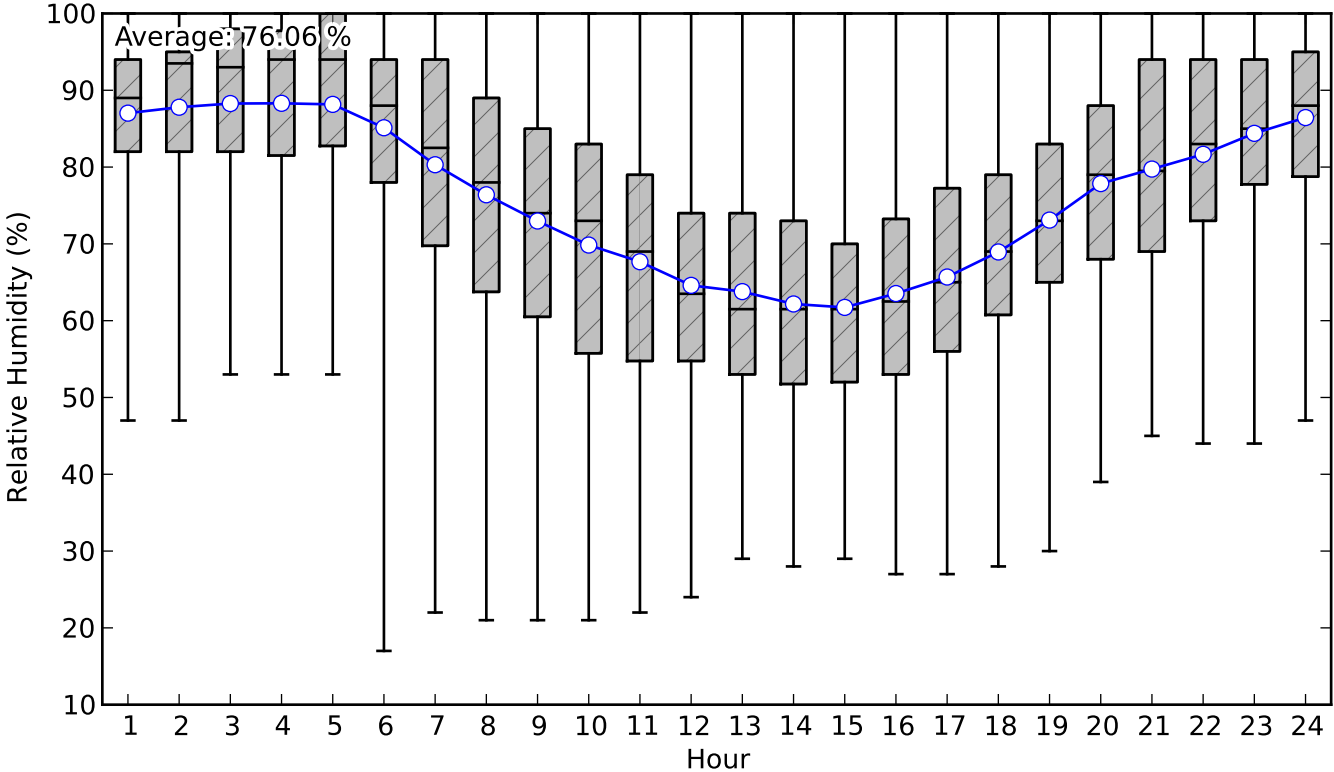
Figure 19: Hourly dewpoint temperature for each season.



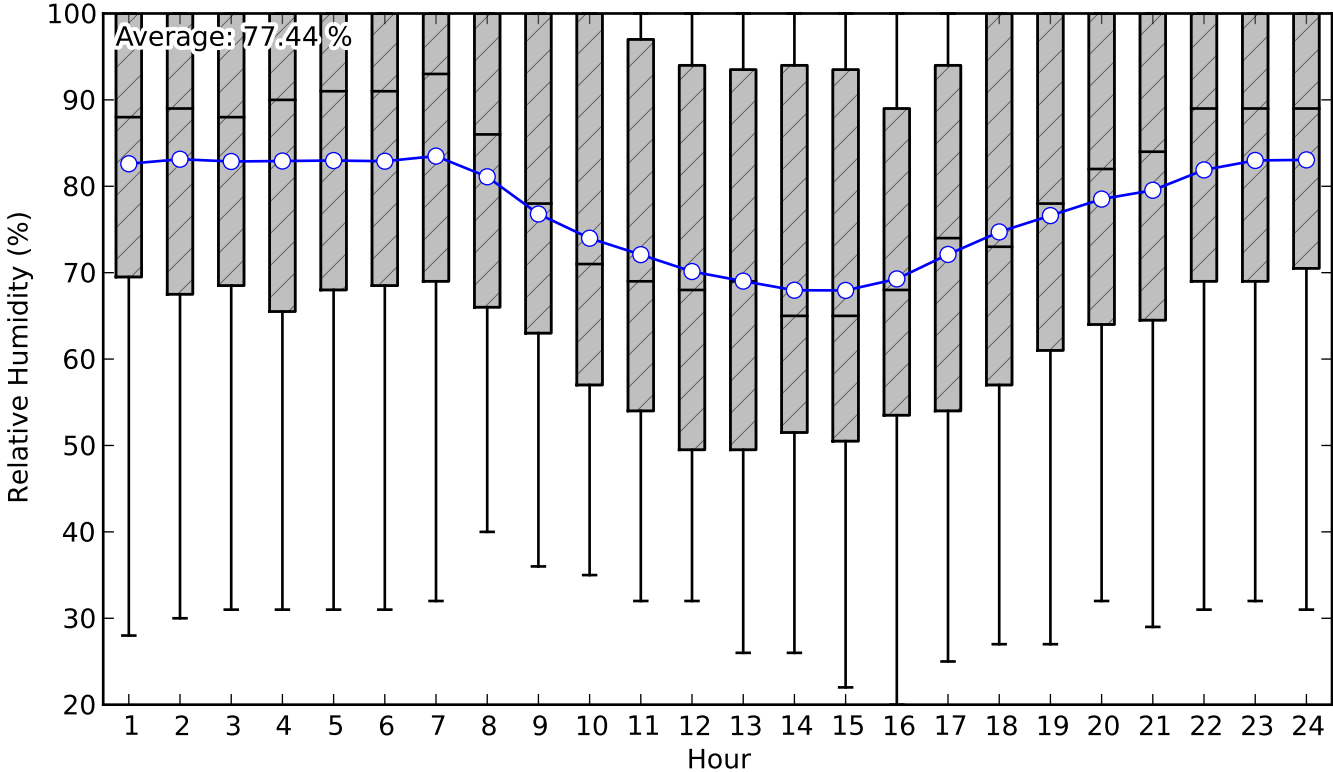
(a) Winter



(b) Spring

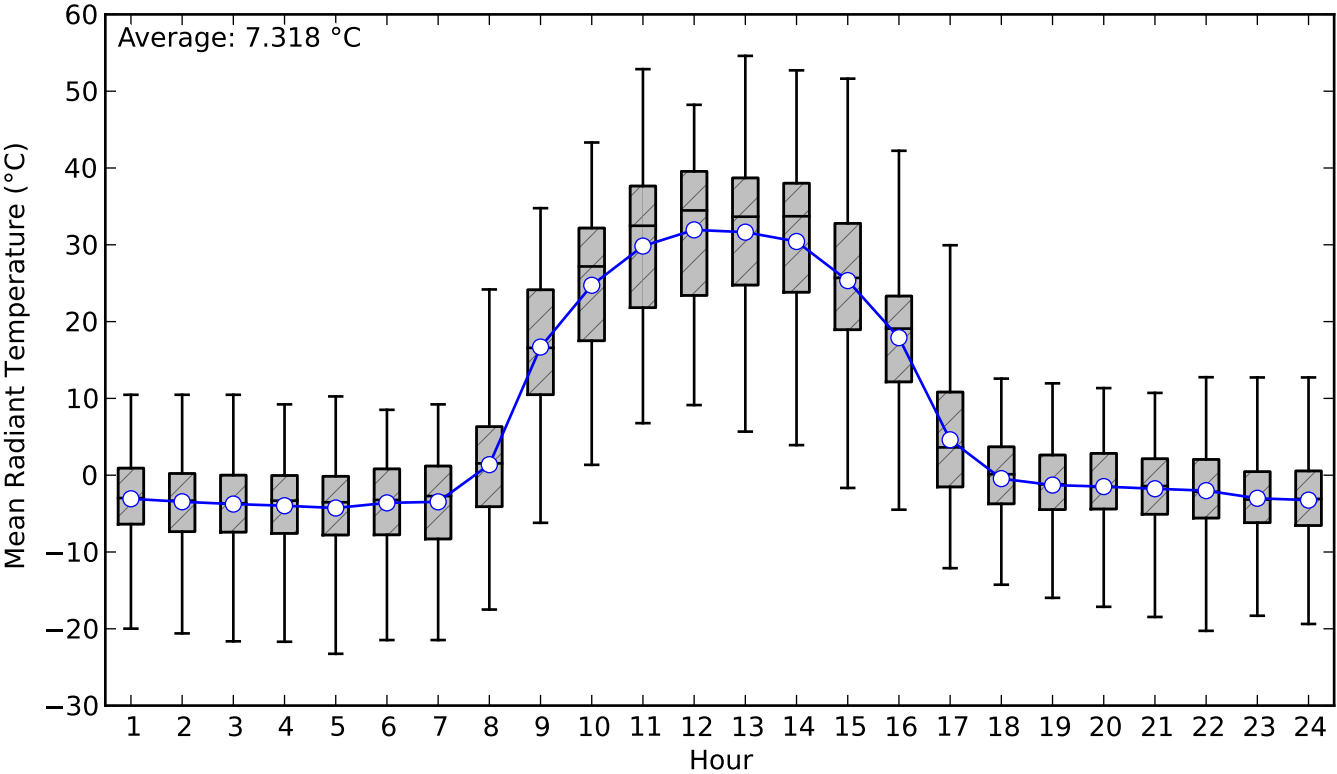


(c) Summer

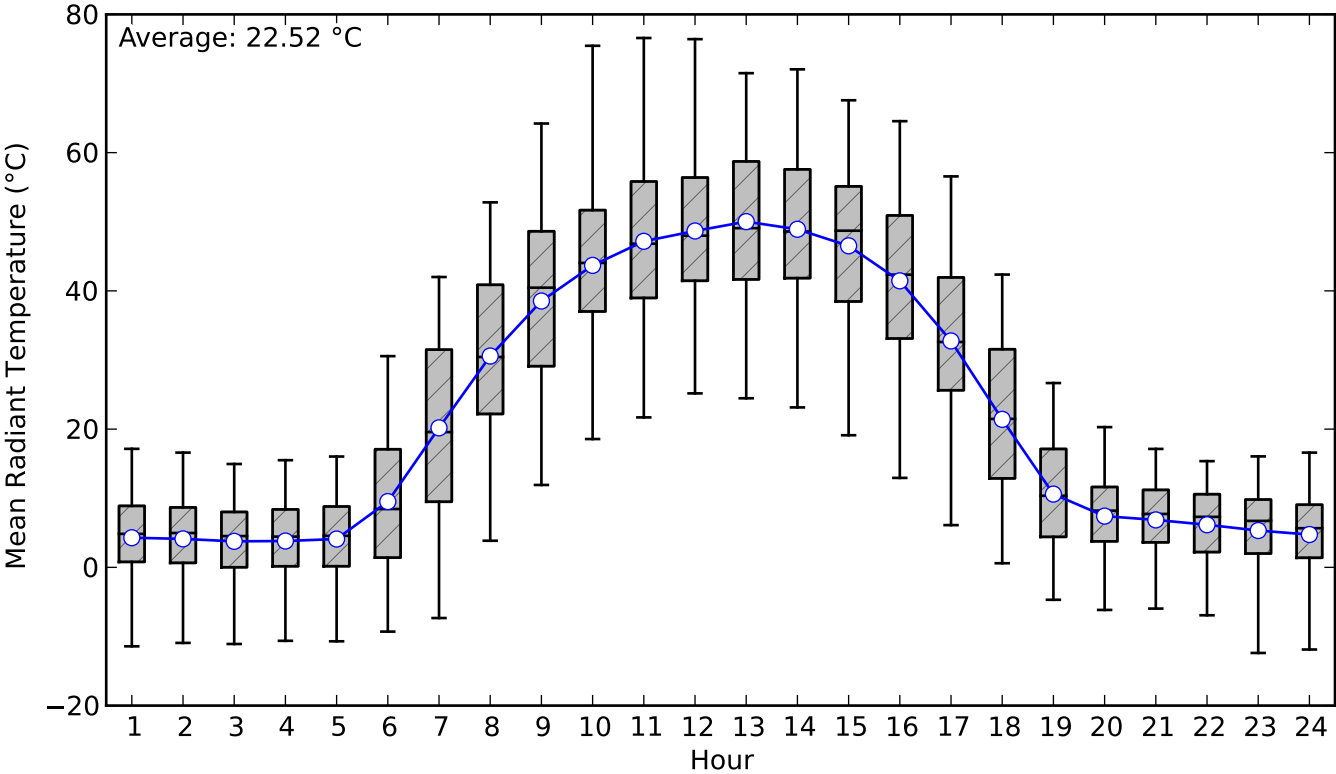


(d) Autumn

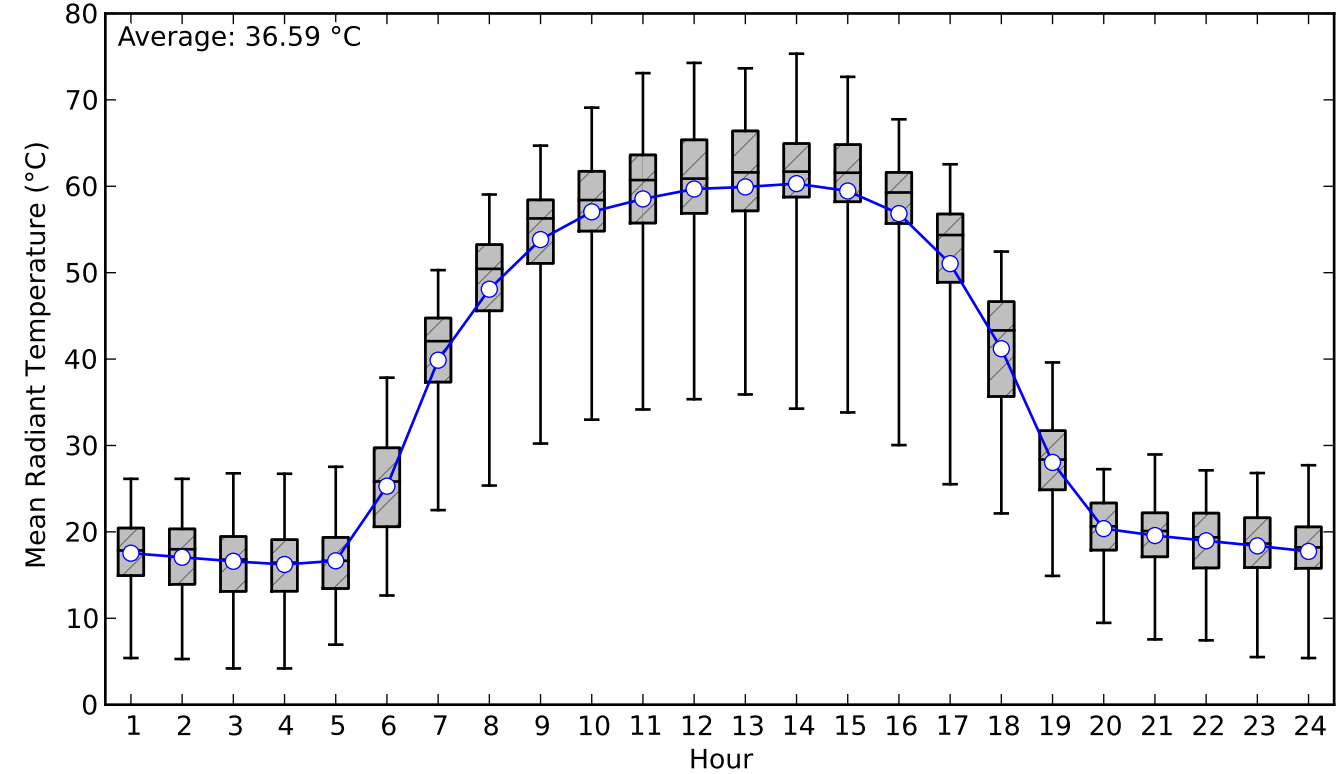
Figure 20: Hourly relative humidity for each season.



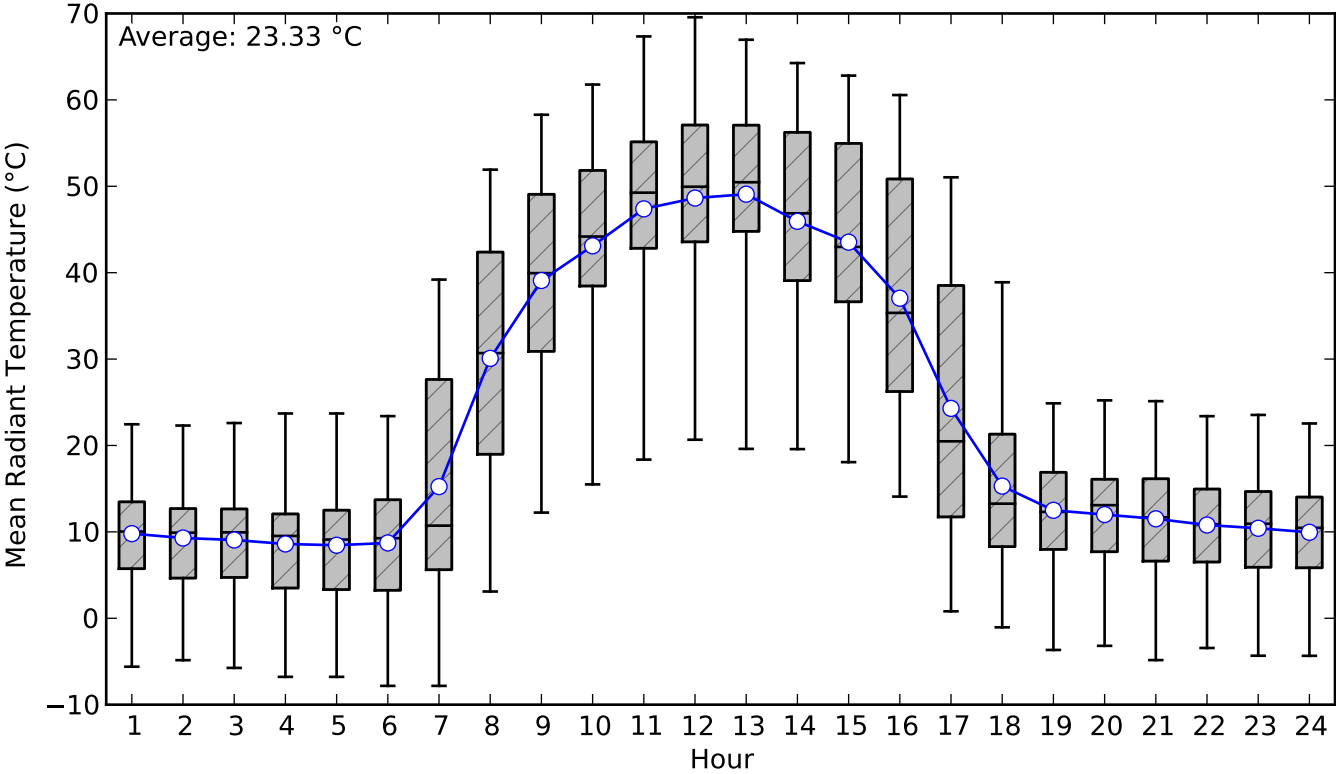
(a) Winter



(b) Spring

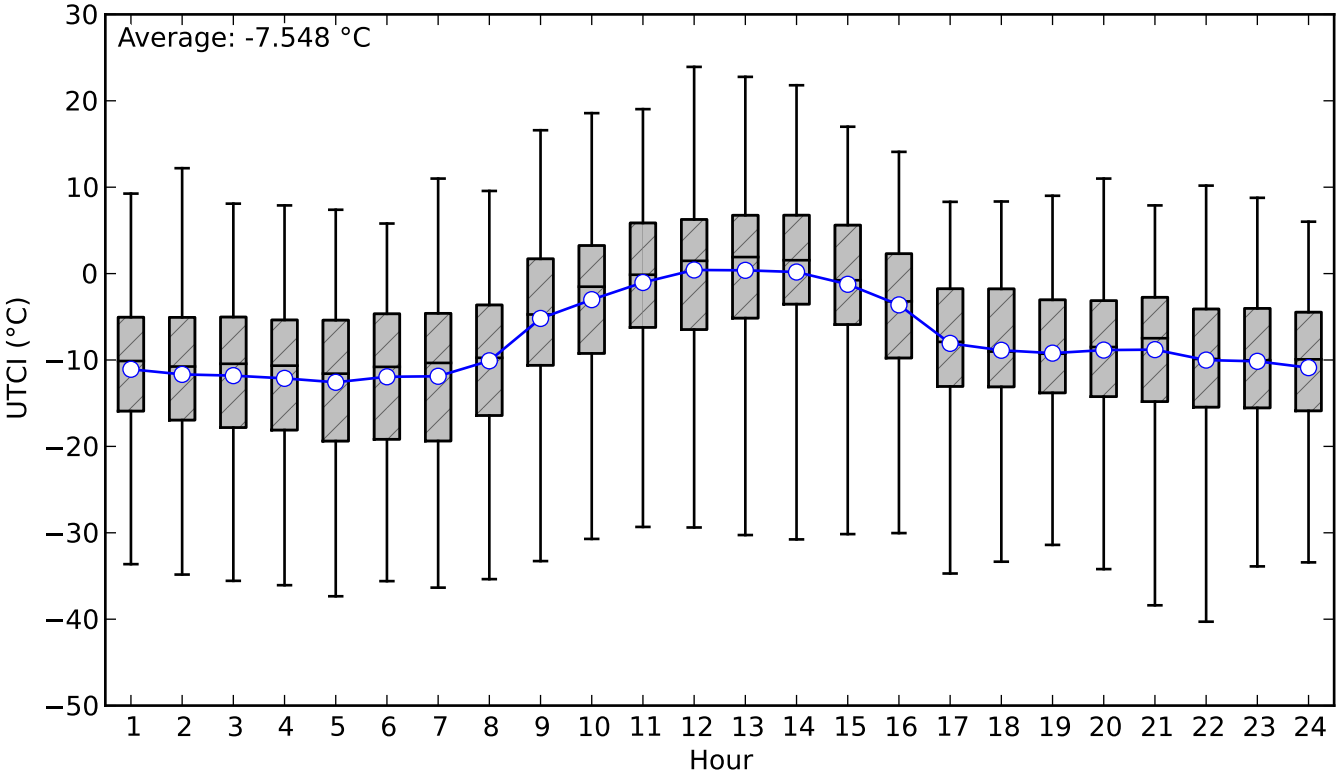


(c) Summer

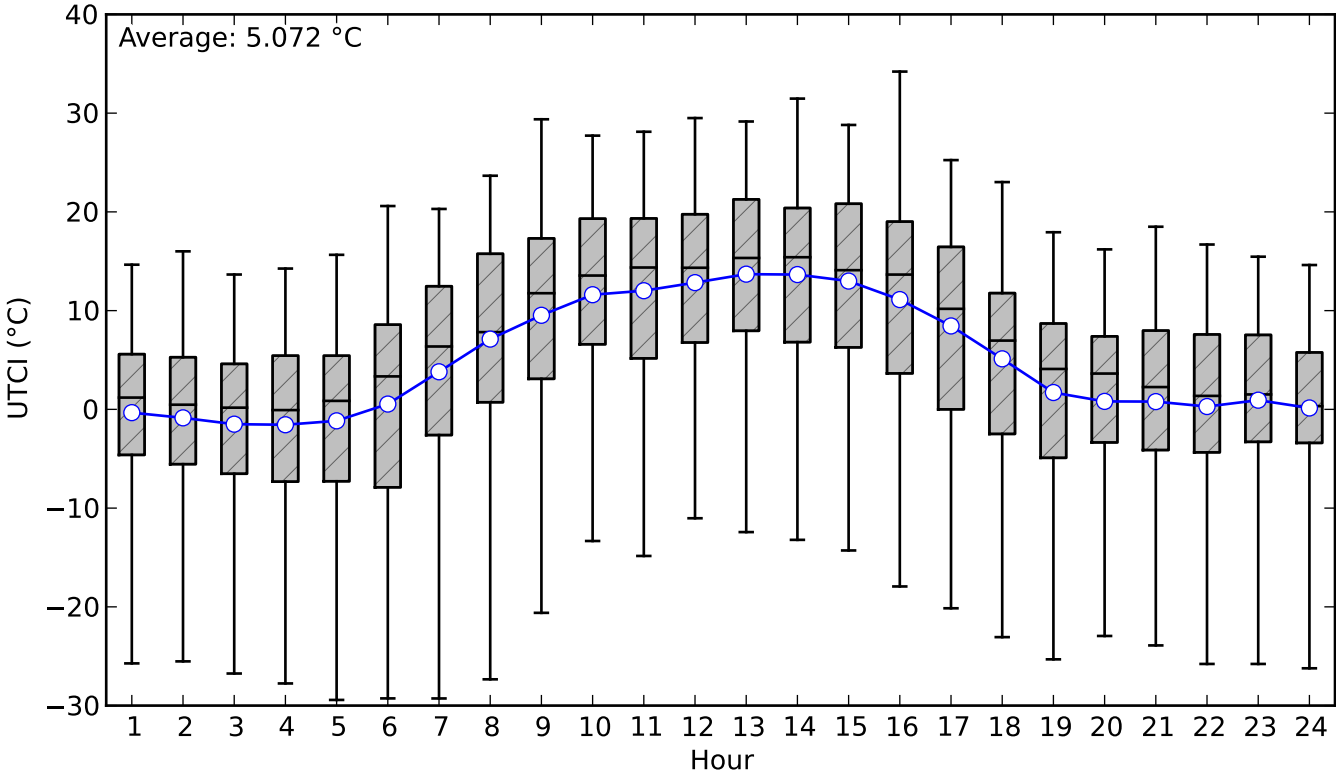


(d) Autumn

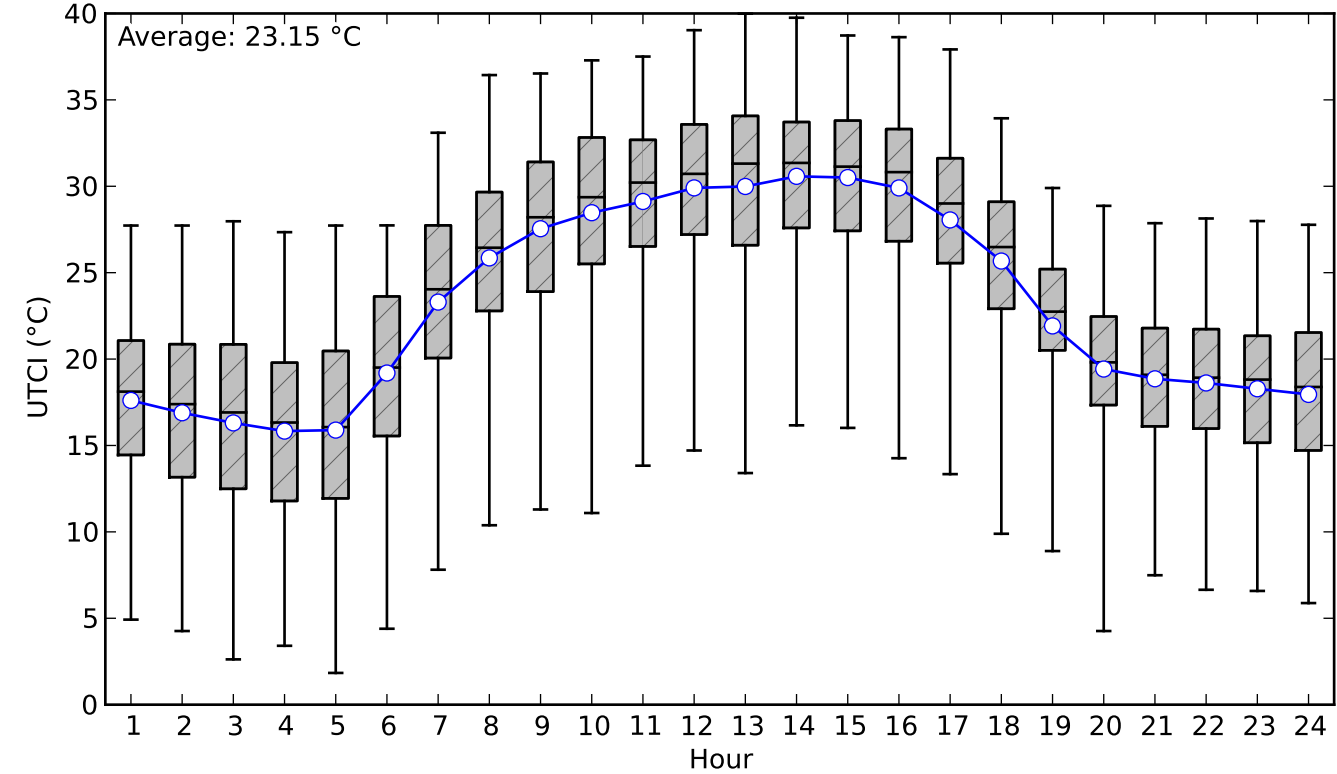
Figure 21: Hourly mean radiant temperature for each season.



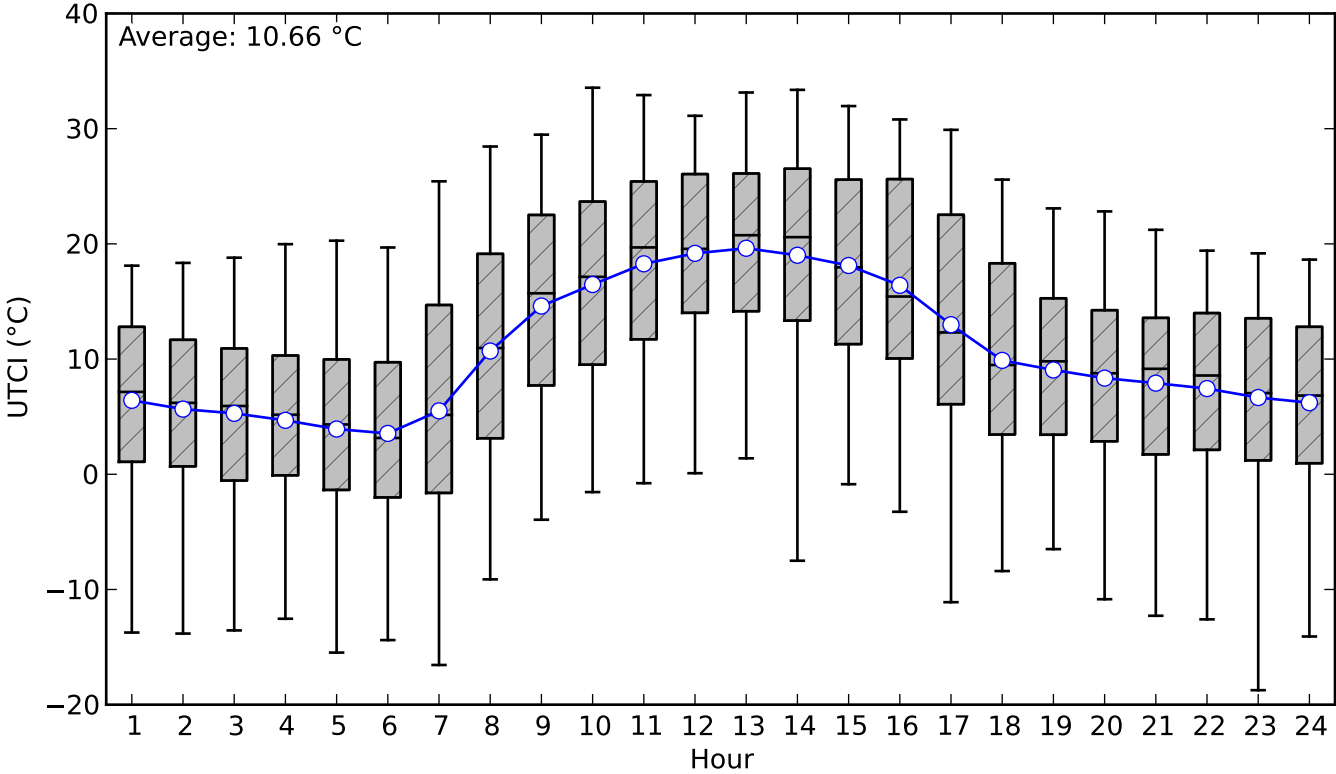
(a) Winter



(b) Spring



(c) Summer



(d) Autumn

Figure 22: Hourly Universal Thermal Climate Index for each season.

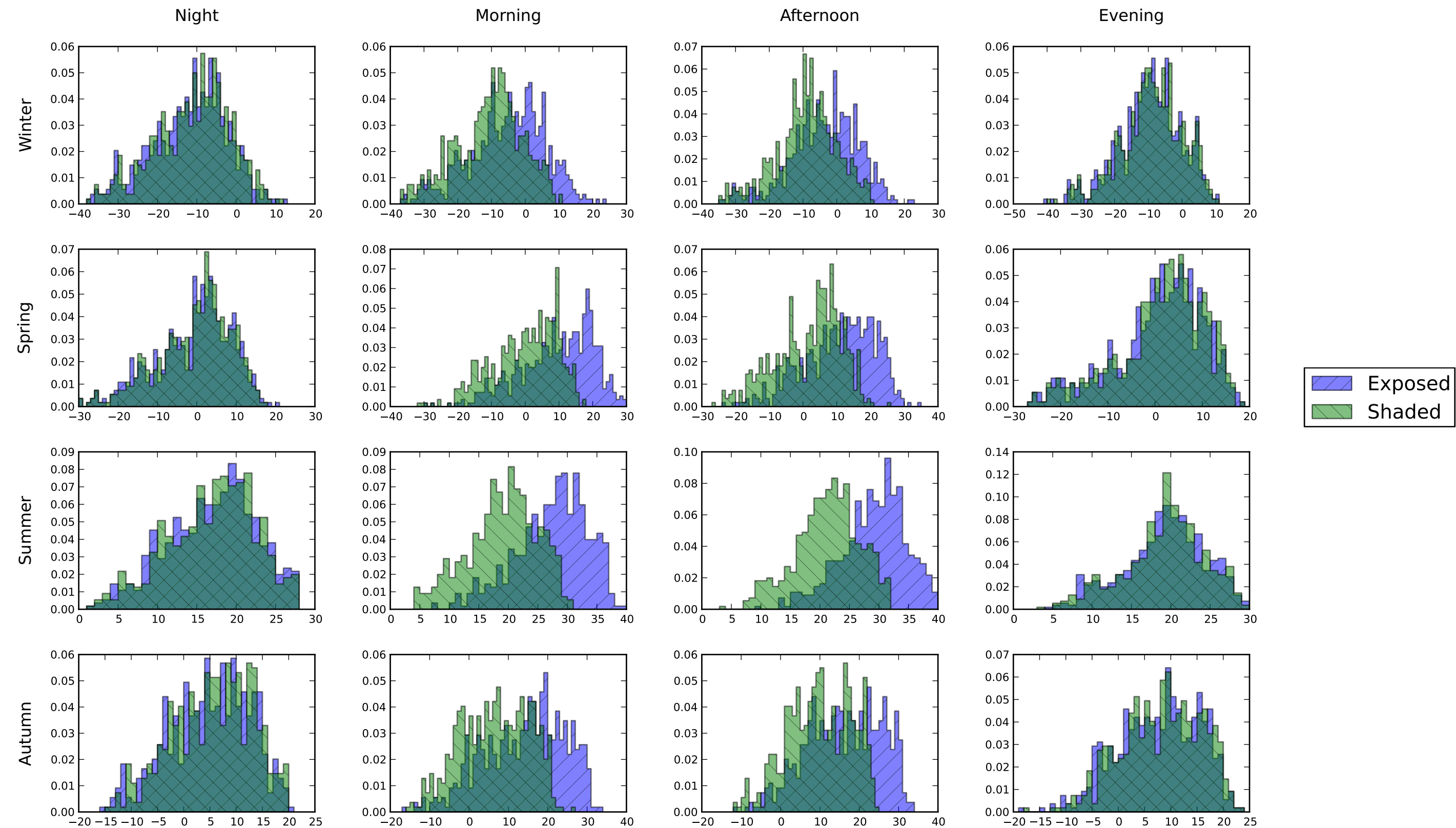


Figure 23: Comparison of frequency of thermal comfort (UTCI) with and without shading for each season and time of day.

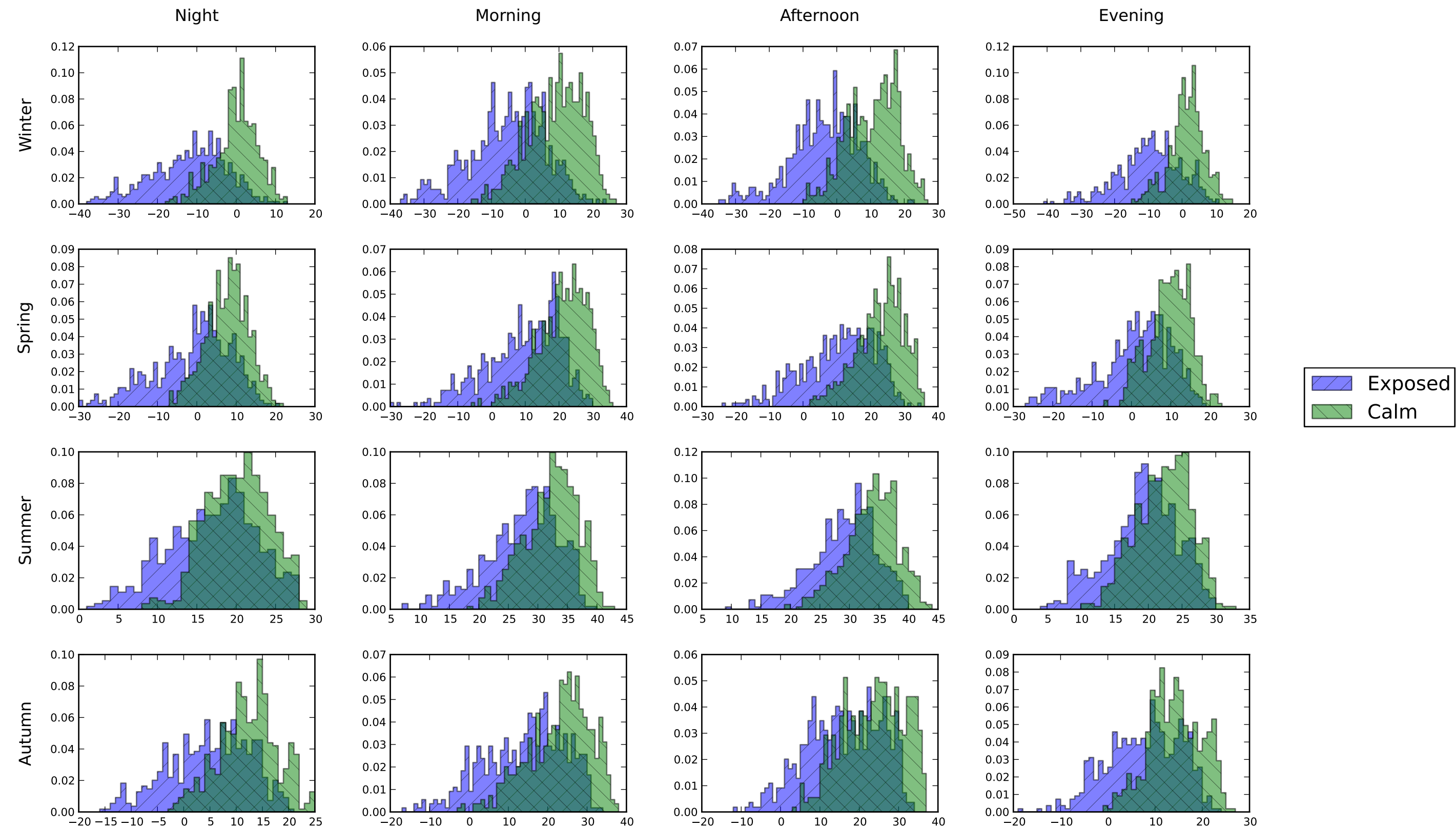


Figure 24: Comparison of frequency of thermal comfort (UTCI) with and without wind for each season and time of day.

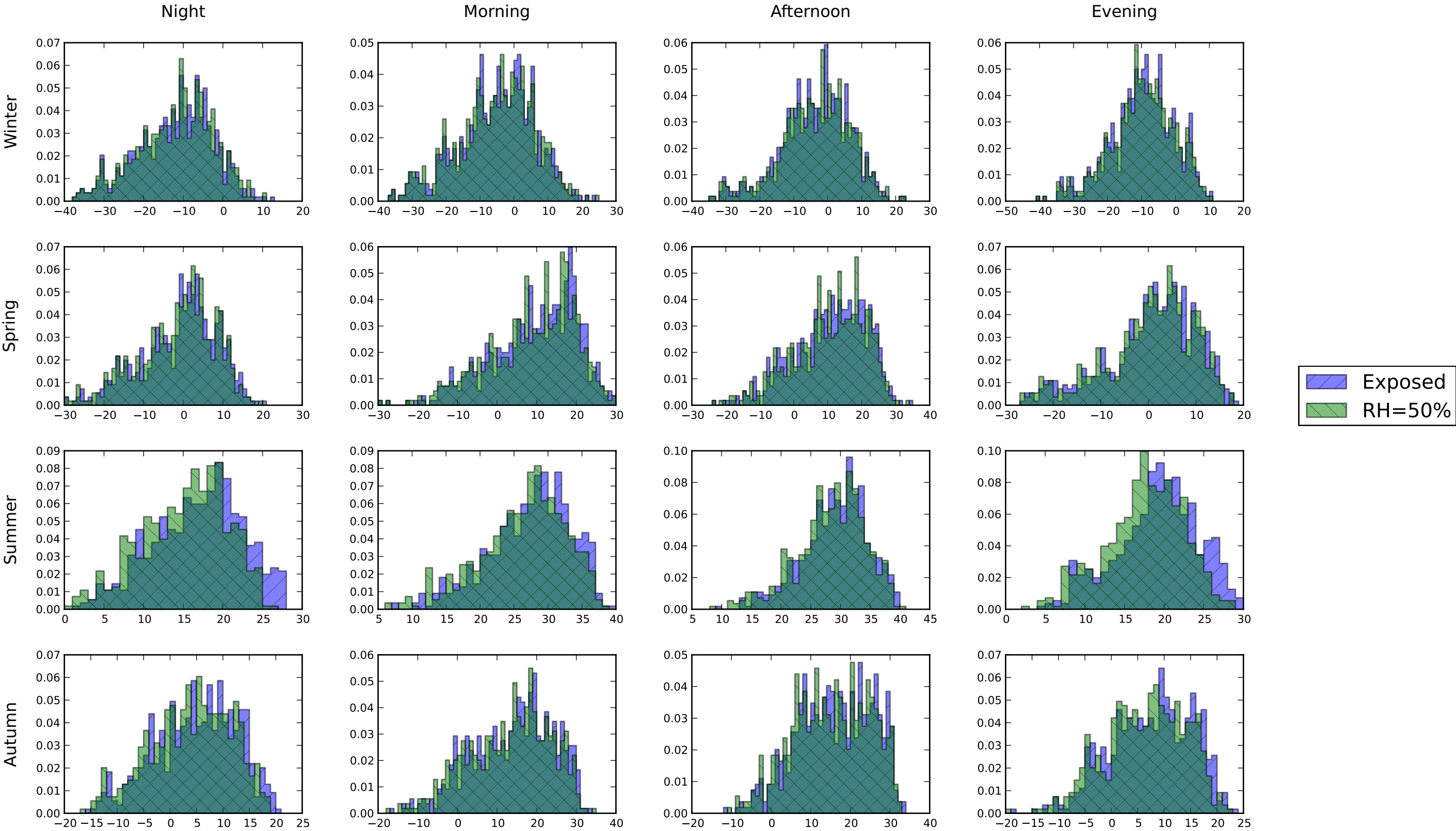
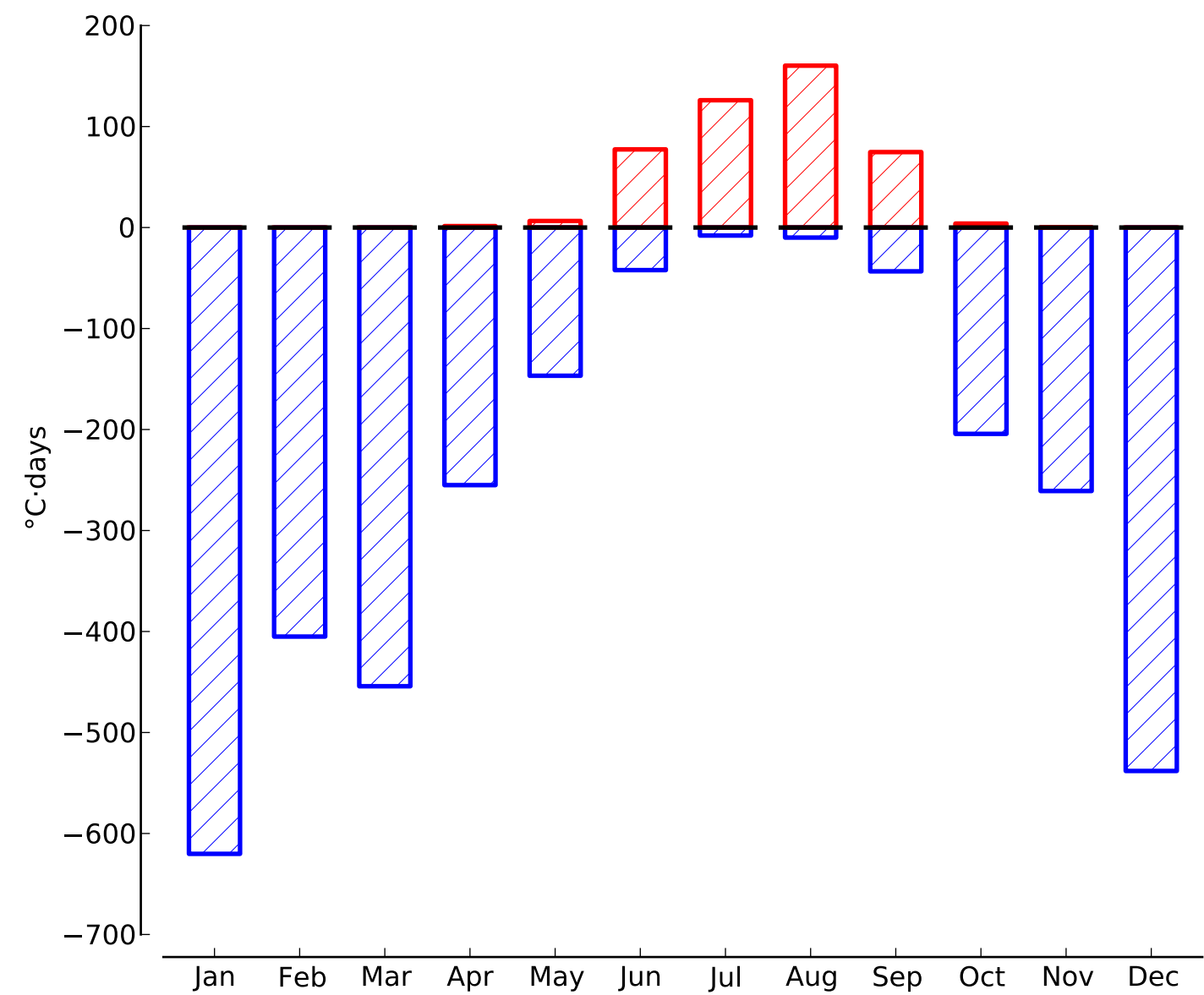
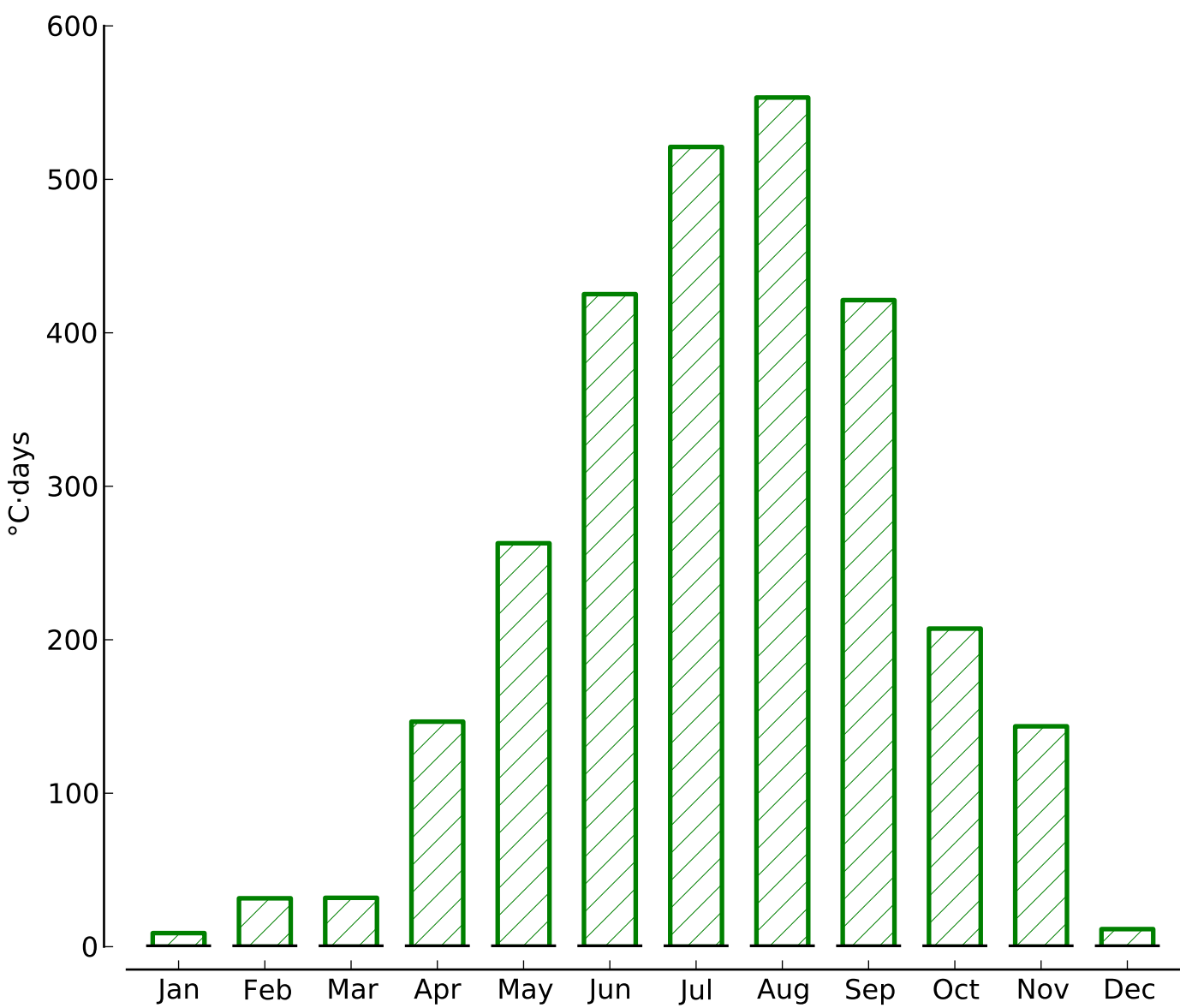


Figure 25: Comparison of frequency of thermal comfort (UTCI) with and without humidity control (RH=50%) for each season and time of day.



(a) Heating and cooling degree days



(b) Growing degree days

Figure 26: Heating (blue) and cooling (red) degree days as calculated against a baseline of 18°C. There are a total of 2987 heat degrees days and 450 cooling degree days. Growing degree days as calculated against a baseline of 5°C. There are a total of 557 growing degree days.

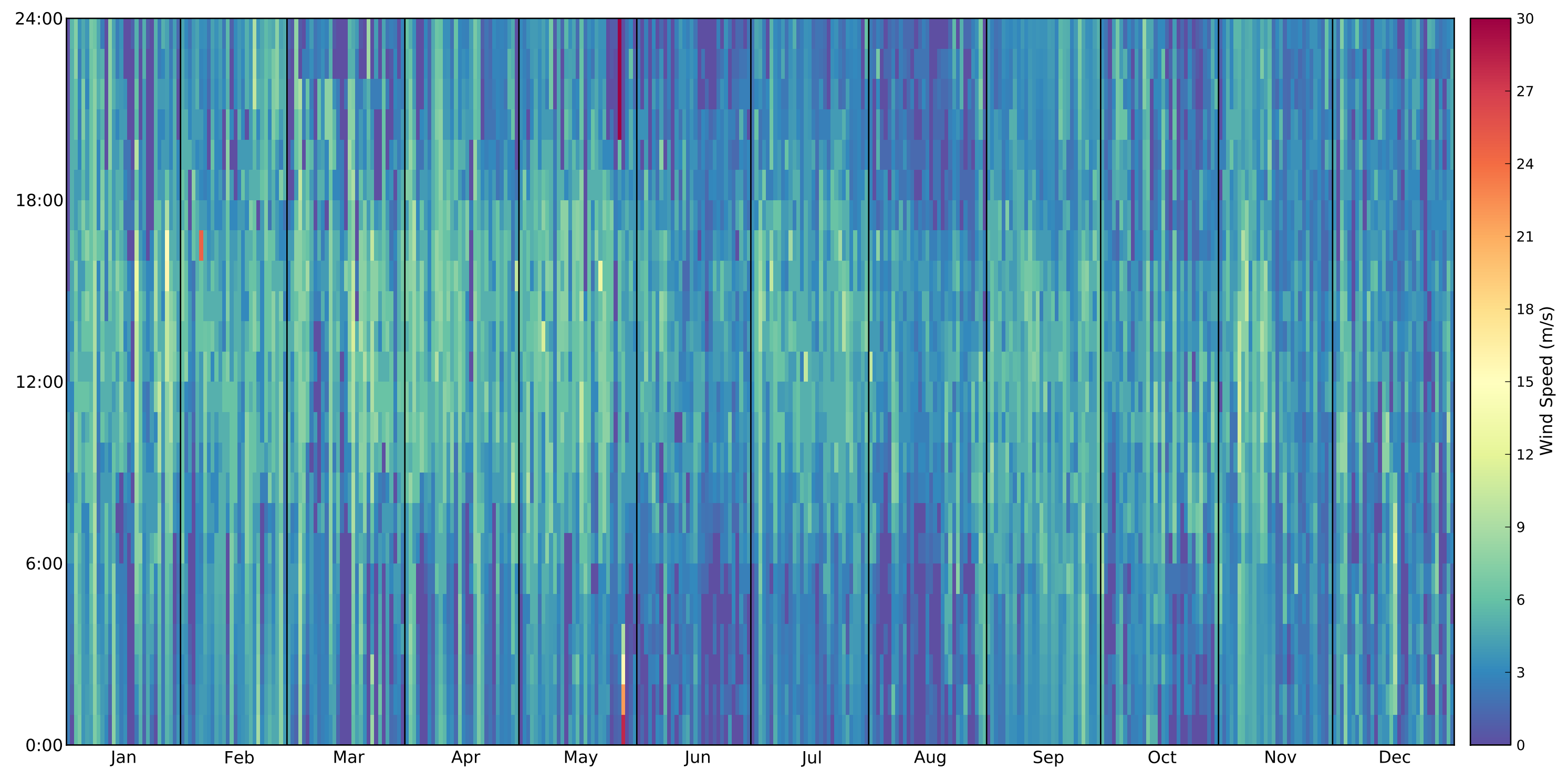


Figure 27: Wind speed for a given day of year and hour of day.

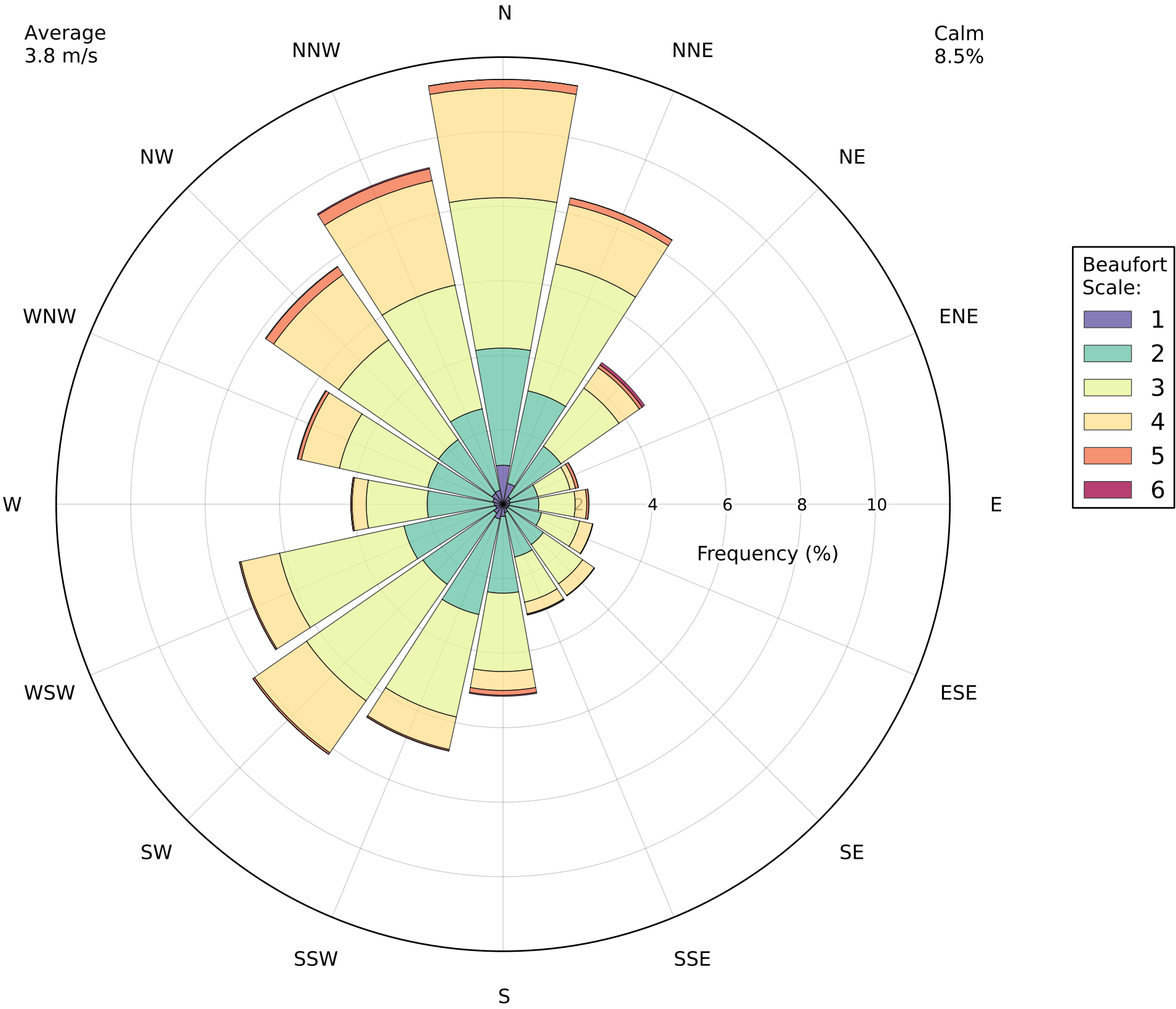


Figure 28: Annual frequency of wind direction and speed. Wind speed bins are based on the [Beaufort Scale](#).

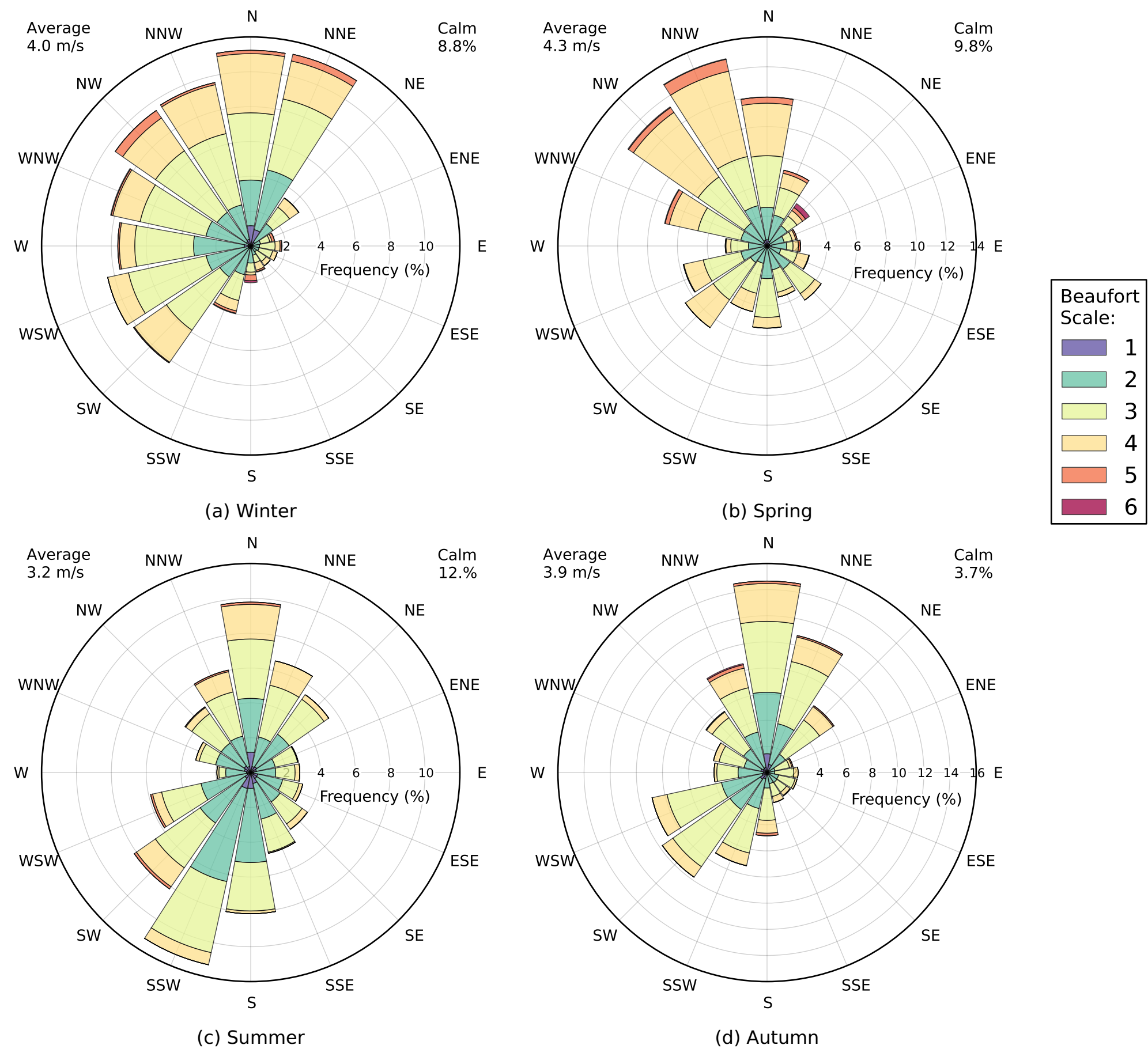


Figure 29: Frequency of wind direction and speed given season.

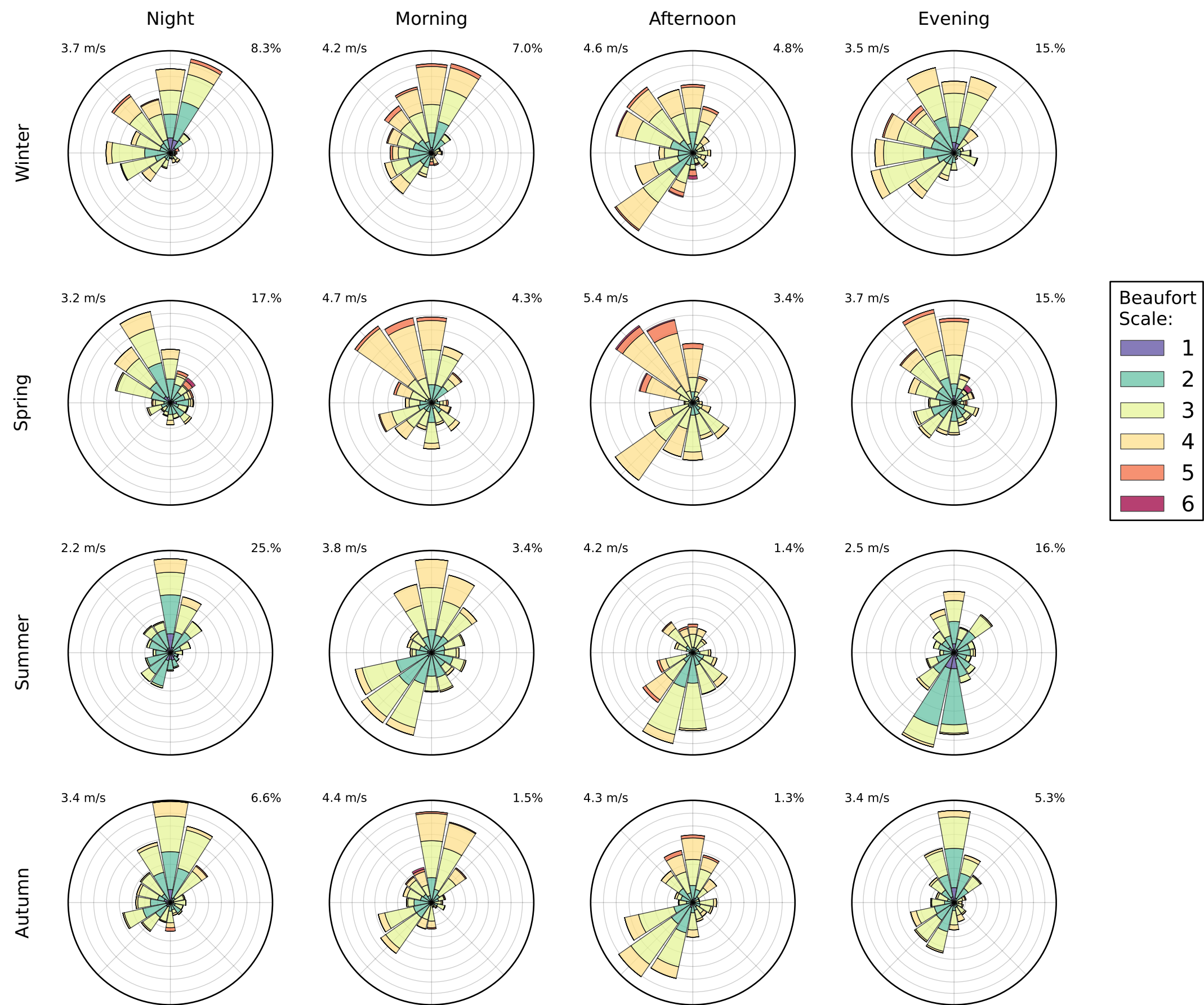


Figure 30: Frequency of wind direction and speed given season and time of day.

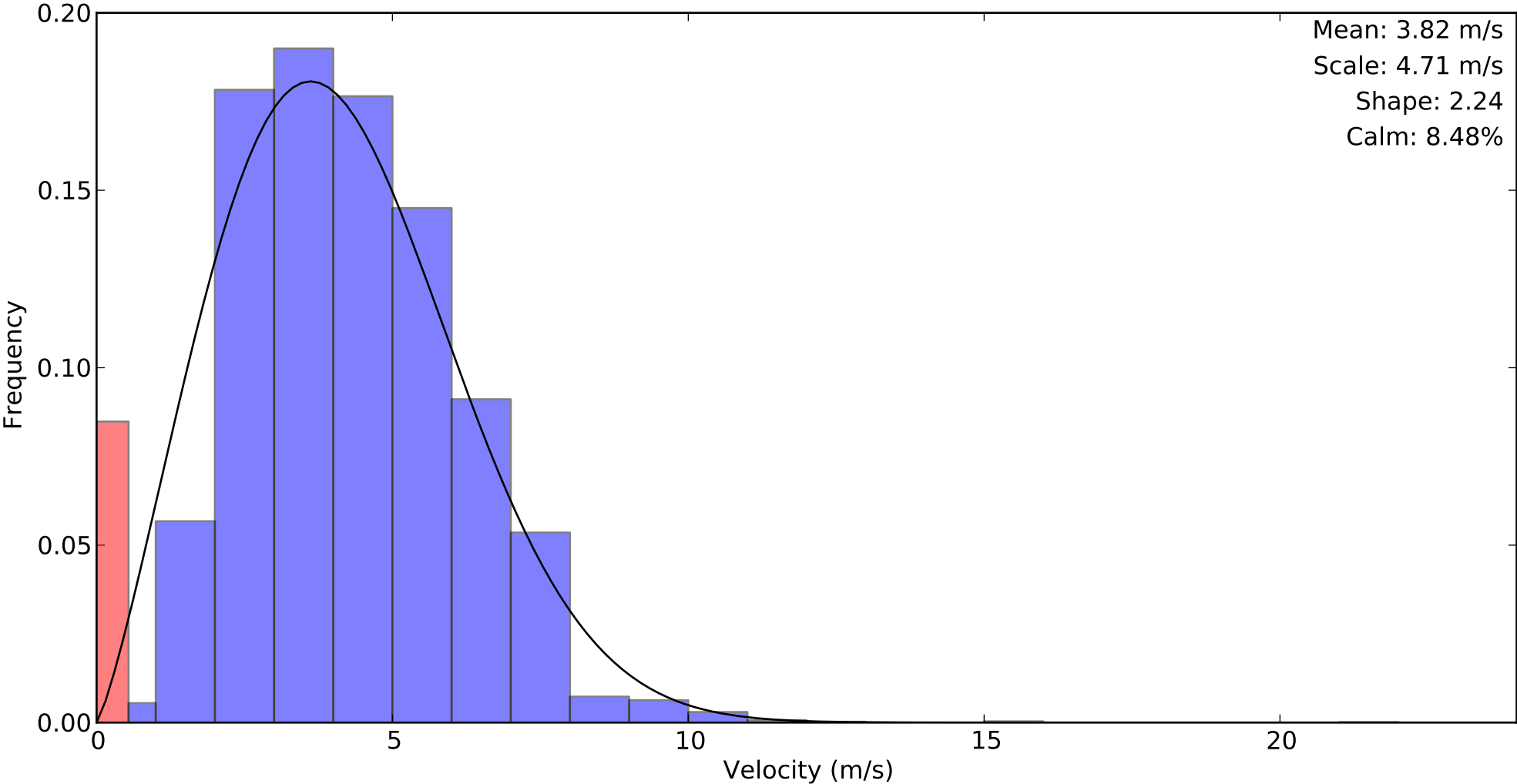


Figure 31: Frequency of wind speed. The black line represents a maximum likelihood estimate of the Weibull distribution for the data. The red bar indicates the frequency of calm.

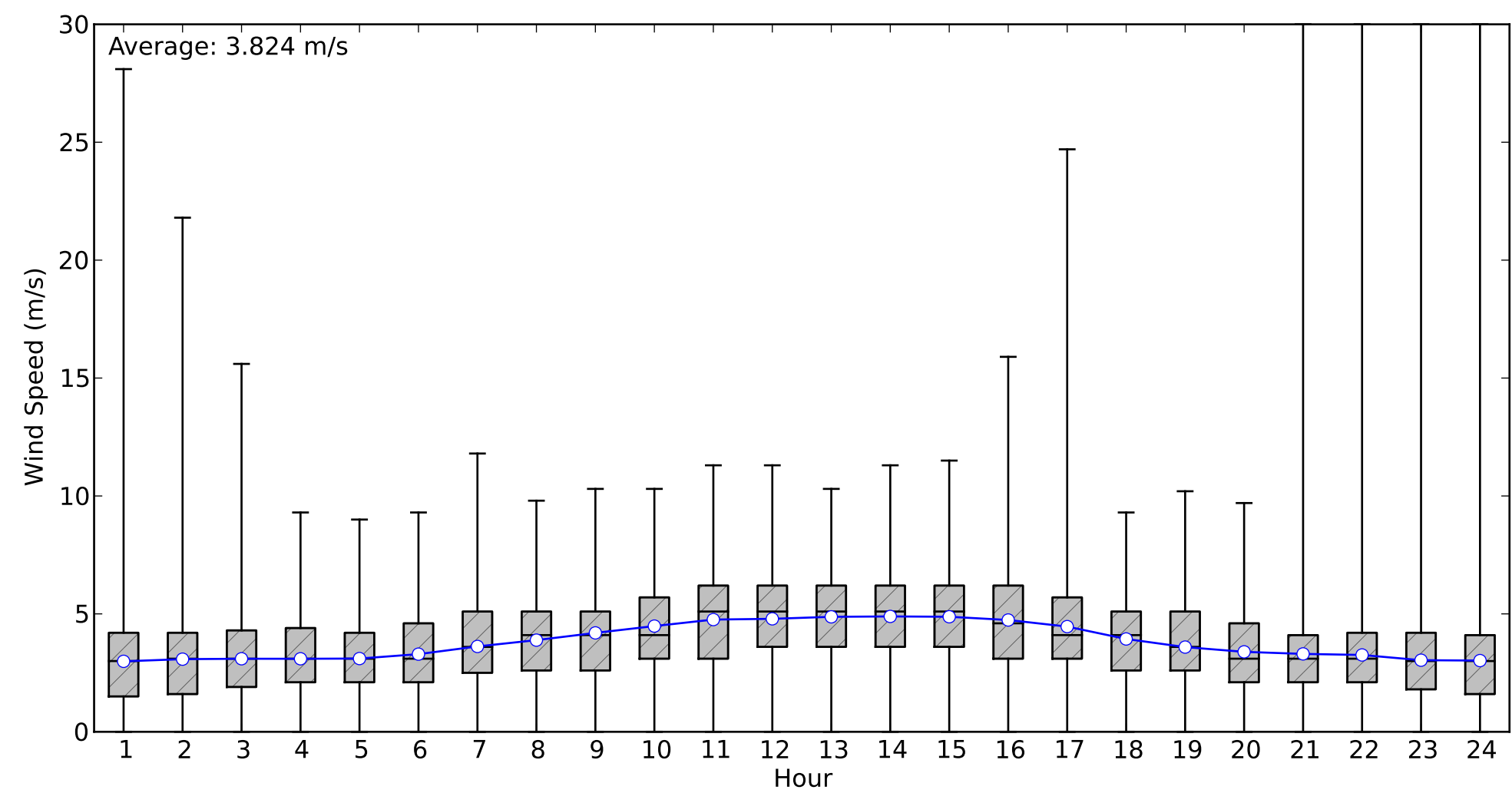
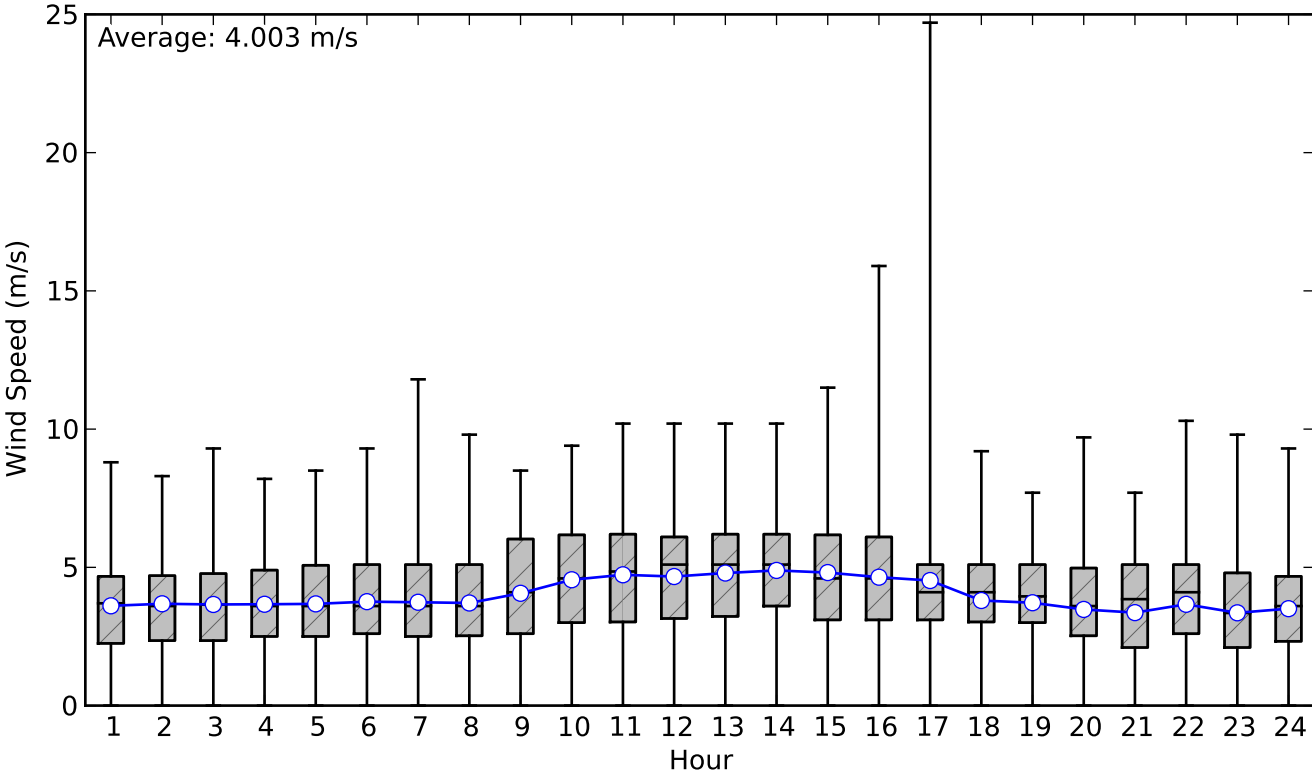
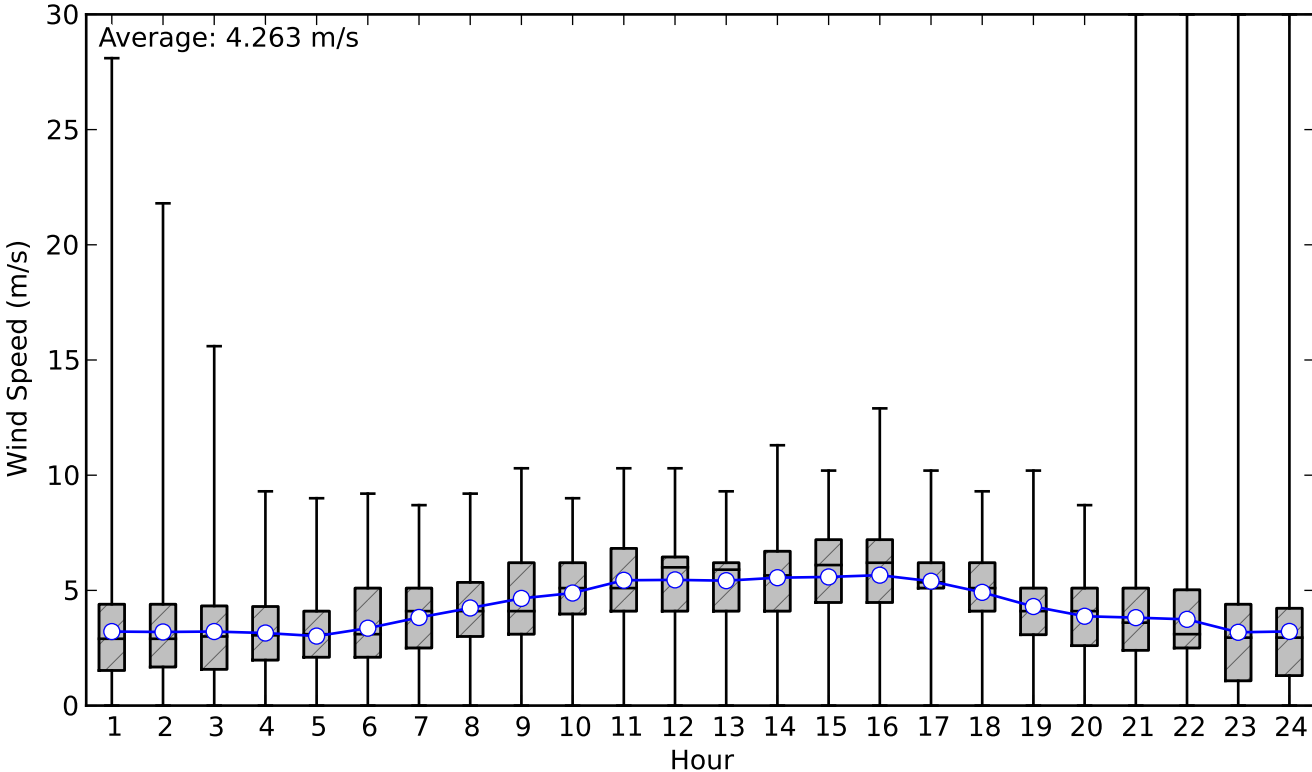


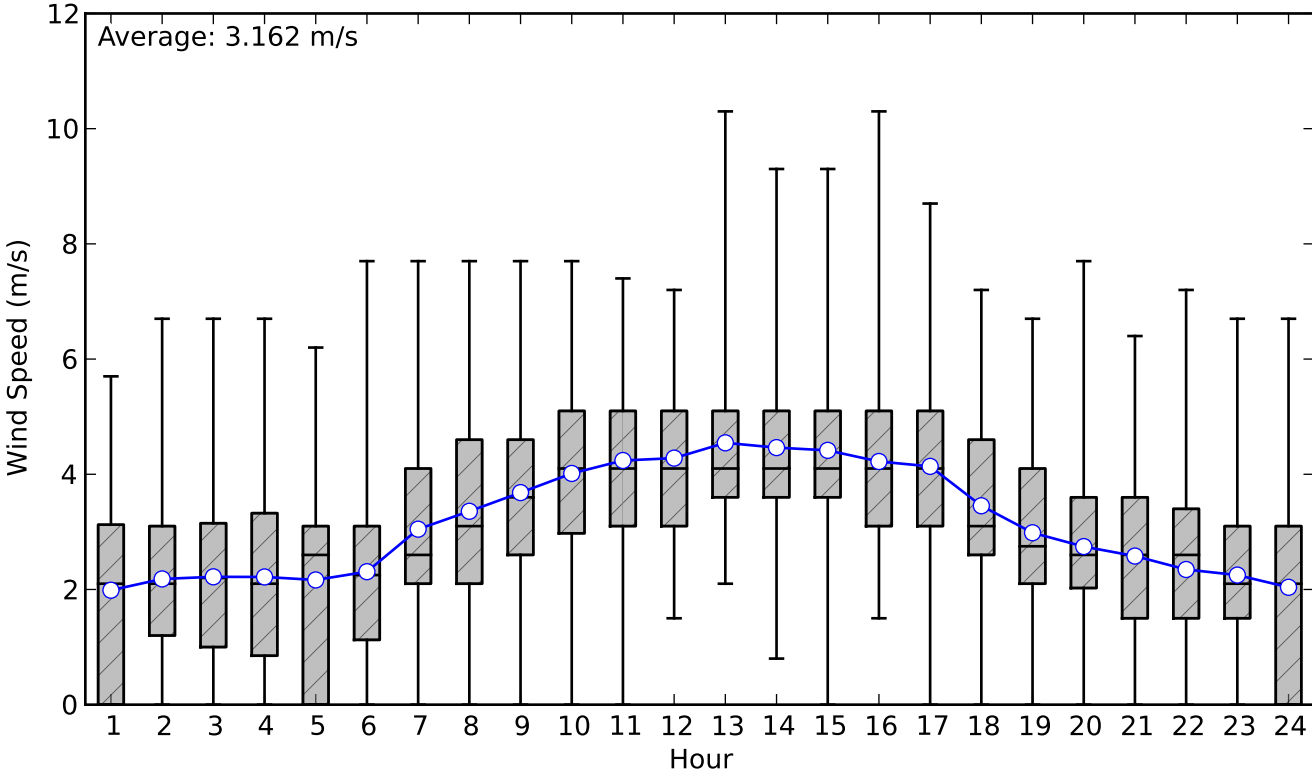
Figure 32: Hourly wind speed for entire year.



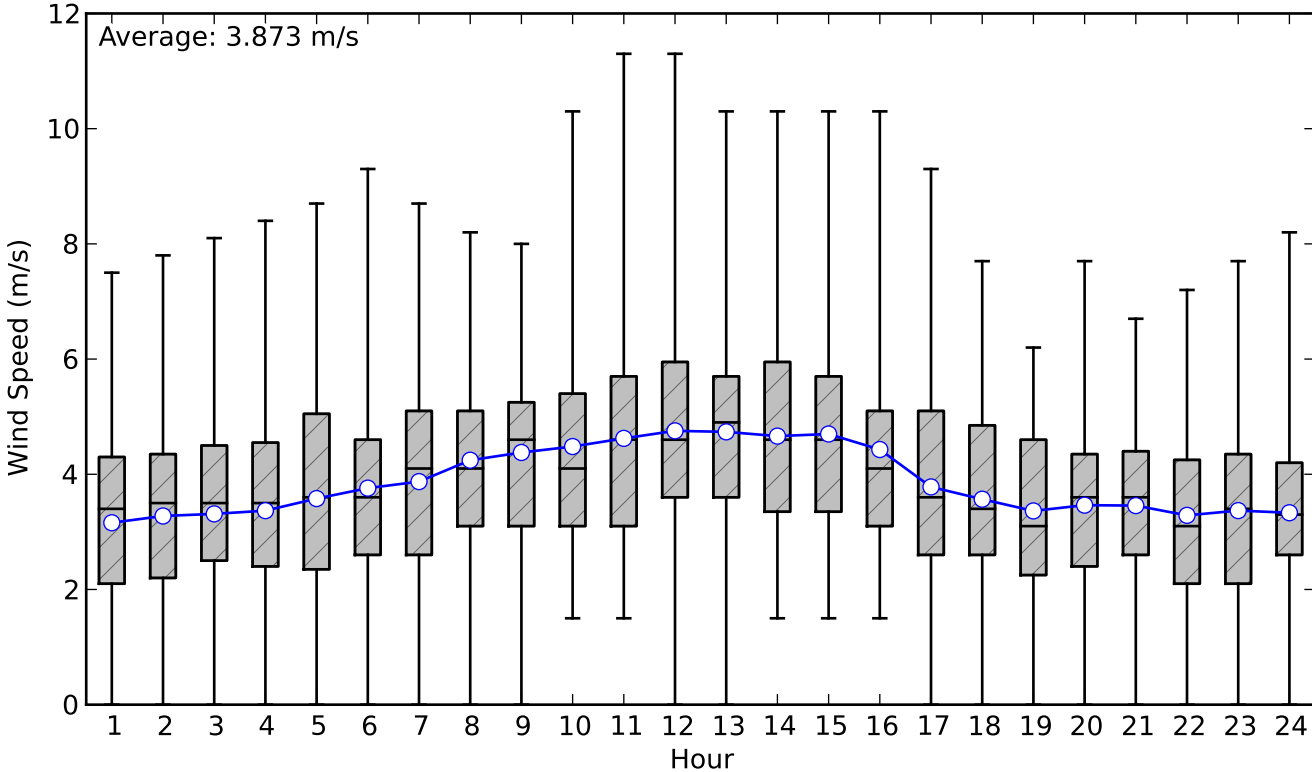
(a) Winter



(b) Spring



(c) Summer



(d) Autumn

Figure 33: Hourly wind speed for each season.

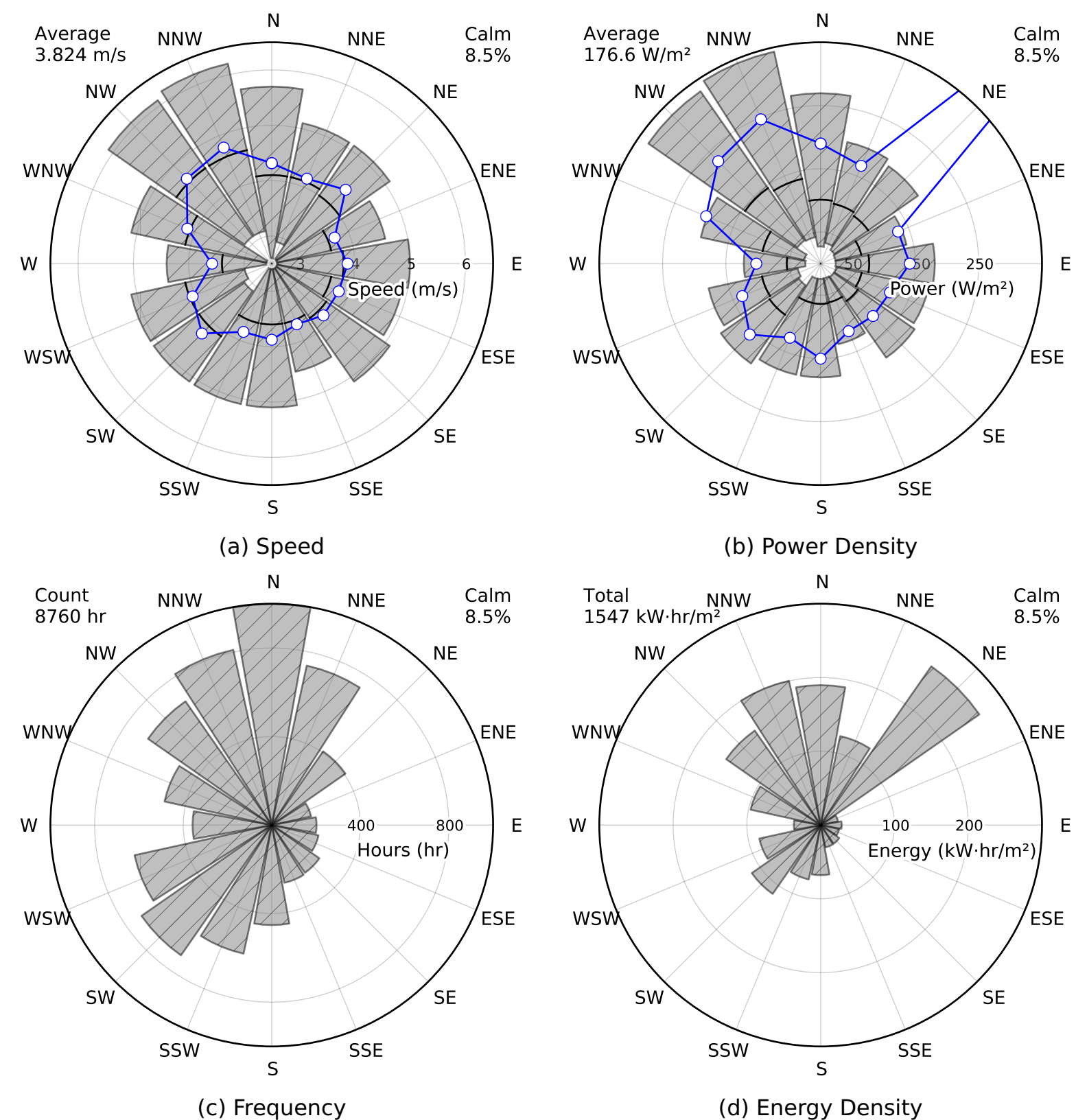


Figure 34: Annual wind speed, wind direction frequency, Wind Power, and total wind energy for a given wind direction at 50m elevation.

Notes

Psychrometric Chart

The psychrometric chart is a useful tool to examine the joint occurrence of temperature and humidity in moist air for accessing comfort or estimating cooling loads.

See <http://en.wikipedia.org/wiki/Psychrometrics> for more information.

Beaufort Scale

The Beaufort Scale bins wind speed data into classes of increasing intensity. A Beaufort of 2 and above typically indicates sufficient wind movement for flushing minor pollutant sources.

Beaufort No.	0	1	2	3	4	5	6
Description	Calm	Light air	Light breeze	Gentle breeze	Moderate breeze	Fresh breeze	Strong breeze
Range (knts)	<1	1-2	3-6	7-10	11-15	16-20	21-26
Range (m/s)	<0.5	0.5-1.6	1.6-3.4	3.4-5.5	5.5-8.0	8.0-10.8	10.8-13.9

See http://en.wikipedia.org/wiki/Beaufort_scale for more information.

Wind Power Classes

Analogous to the Beaufort Scale, one can classify the wind power density.

Power Class	1	2	3	4	6	6	7
Description	Poor	Marginal	Fair	Good	Excellent	Outstanding	Superb
Range (W/m²)	<200	200-300	300-400	400-500	500-600	600-800	800-1600

See http://en.wikipedia.org/wiki/Wind_power for more information.

Universal Thermal Climate Index

Universal Thermal Comfort Index (UTCI). UTCI is similar to other adaptive models of thermal comfort in that it relates temperature, humidity, radiative exposure, and wind speed to a perceived temperature. If the baseline conditions are 20°C, a relative humidity of 50%, still air, and no radiative difference between surroundings, e.g. walls, sky, ground temperatures, solar impact, grouped into a mean radiative temperature (MRT), the resulting UTCI is 20°C.

As a rule of thumb, an increase in relative humidity of 20% raises UTCI, or the perception of temperature, by 1°C. Likewise, an increase in MRT of 3°C results in an increase in UTCI of 1°C. The impact of wind speed is non-linear, with a rapidly increasing cooling benefit resulting from increasing wind speed: increasing the wind speed by 1.3 m/s results in an perceived cooling of 1°C but an increase of 2 m/s results in a perceived cooling of 2°C.

See <http://www.utci.org> for more information.

Weibull Distribution

The Weibull distribution is a probability density distribution defined as

$$f(x;\lambda,k) = \frac{k}{\lambda} \left(\frac{x}{\lambda}\right)^{k-1} \exp\left(-\left(\frac{x}{\lambda}\right)^k\right)$$

where k is the “shape parameter” and λ is the scale parameter.

See http://en.wikipedia.org/wiki/Weibull_distribution for more information.

Typical Meteorological Year

This information was auto-generated from information based on a single year of data assembled as representative of a “Typical Meteorological Year” or TMY. This data may or may not be consistent with the long-term averages of the location in question. This data is not suitable for extreme value analysis.

See http://en.wikipedia.org/wiki/Typical_meteorological_year for more information.

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